

SYSTEMATICS AND PHYLOGENY

Systematics of the genus *Geastrum* (Fungi: Basidiomycota) revisitedJuan Carlos Zamora,¹ Francisco de Diego Calonge,¹ Kentaro Hosaka² & María Paz Martín¹¹ Real Jardín Botánico – CSIC, Plaza de Murillo 2, 28014 Madrid, Spain² Department of Botany, National Museum of Nature and Science, 4-1-1 Amakubo, Tsukuba, Ibaraki 305-0005, Japan

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Abstract Various infrageneric classifications of the genus *Geastrum* have been proposed, all of them exclusively based on morphology. They are highly divergent due to the different relative value that each author gave to particular morphological features. There is no consensus about any of these classifications, and therefore most modern studies avoid classifying *Geastrum* species in supraspecific units. Here, a new subdivision of the genus *Geastrum* is presented, based on phylogenetic analyses of four molecular markers, viz., 5.8S nrDNA, nrLSU, *rpb1*, and *atp6*. This classification is also supported by chemical reactions with syringaldazine and guaiac gum, and by some morphological characters. Most reliable morphological characters include both classically studied features (e.g., endoperidial surface, peristome structure, mycelial layer), and recently discovered ones (e.g., rhizomorph crystals). Previously published infrageneric classifications are reviewed, showing many discrepancies with the phylogenetic data. A new infrageneric subdivision is proposed: sect. *Campestris* sect. nov., sect. *Corollina* sect. nov. (including subsect. *Lageniformia* subsect. nov., subsect. *Marginata*, and subsect. *Plicostomata*), sect. *Elegantia* sect. nov., sect. *Exareolata*, sect. *Fimbriata*, sect. *Fornicata*, sect. *Geastrum* (including subsect. *Arenaria* subsect. nov., subsect. *Geastrum*, subsect. *Quadrifida* subsect. nov., and subsect. *Sulcostomata*), sect. *Hariotia* sect. nov., sect. *Hieronymia* sect. nov., sect. *Myceliostroma* (including subsect. *Epigaea* and subsect. *Velutina* subsect. nov.), sect. *Papillata*, sect. *Pseudolimbata* sect. nov., sect. *Schmidelia* sect. nov., and sect. *Trichaster*. The phylogenetic and morphological boundaries among three genera of Geastraceae, viz., *Geastrum*, *Myriostoma*, and *Radiigera*, are also evaluated. *Myriostoma* likely represents a different phylogenetic lineage within Geastraceae and is confirmed as distinct from *Geastrum*. However, *Radiigera* is nested in *Geastrum*, and therefore it is confirmed as a synonym of *Geastrum*.

Keywords 5.8S; *atp6*; endoperidial surfaces; Geastraceae; nomenclature; nrLSU; phenoloxidase spot tests; rhizomorph crystals; *rpb1*; taxonomy

■ INTRODUCTION

The genus *Geastrum* Pers.:Pers., widely known as the earthstars, includes those species of gasteroid Basidiomycota with a complex peridium, which is anatomically pluristratified and functionally divided in one to three layers, the exoperidium, mesoperidium, and endoperidium (Sunhede, 1989). The exoperidium normally splits at maturity into a characteristic star-like structure, the endoperidium encloses the fertile glebal mass in most species (Staněk, 1958) and, when present, the mesoperidium remains attached to both the inner exoperidial surface and the outer endoperidial surface (Sunhede, 1989) (Fig. 1). The gleba of immature basidiomata is organized in chambers covered by a true hymenium, basidia are typically multispored, and generative hyphae are provided with clamp connections at primary septa. A true capillitium is present in the mature gleba, and basidiospores are double-layered under the transmission electron microscope, with a single-walled epispore, symmetrically attached to the sterigmata, and ornamented with normally cylindrical, baculate or irregularly conical processes (Bronchart & al., 1975; Sunhede, 1989).

The earthstars have a subcosmopolitan distribution. They have been recorded from all continents except Antarctica, and are more abundant in the temperate zones and in the tropics (Ponce de León, 1968). Currently, around 50 species are accepted (Kirk & al., 2008) but, according to our experience from herbarium specimens and field work, 100–120 species may belong to this genus.

Geastrum was firstly proposed by Persoon (1794), and later adopted by himself (Persoon, 1801) in his *Synopsis Methodica Fungorum*. Problems with its orthographic variant *Geaster* P.Micheli ex Fr. (Fries, 1829) and its typification have been carefully treated by Demoulin (1984, but see taxonomic part). Persoon (1794) first included only three species, viz., *G. rufescens*, *G. quadrifidum* and *G. multifidum*, increasing them to six in 1801, of which two (*G. coliforme* With.:Pers. and *G. hygrometricum* Pers.) are currently included in other genera, i.e., *Myriostoma coliforme* (With.:Pers.) Corda and *Astraeus hygrometricus* (Pers.) Morgan. After Persoon (1801) many more taxa have been described, either under *Geaster* or *Geastrum*, resulting in roughly 300 names (Robert & al., 2005). The genus is morphologically well defined, but some species

have also been placed in other genera such as *Coilomyces* Berk. & M.A. Curtis, *Cycloderma* Klotzsch, *Geasteroides* Battarra, *Plecostoma* Desv., and *Trichaster* Czern. All of these genera are currently considered synonyms of *Geastrum* (Fischer, 1933; Staněk, 1958; Ponce de León, 1968; Sunhede, 1989; Kasuya & al., 2012).

Geastrum is the largest genus of Geastraceae Corda which contains, according to Kirk & al. (2008), the following seven genera: *Sphaerobolus* Tode: Pers., *Schenella* T. Macbr., *Radiigera* Zeller, *Myriostoma* Desv., *Geastrum*, *Terrostella* Long (as *Geasteroides* Long), and *Phialastrum* Sunhede. *Sphaerobolus* and *Schenella* currently are considered to belong to other families (Hosaka & Castellano, 2008). Sunhede (1989) considered Geastraceae to include eight genera, viz., *Geasteropsis* Hollós, *Geastrum*, *Myriostoma*, *Phialastrum*, *Radiigera*, *Schenella* (as *Pyrenogaster* Malençon & Rioussset), *Terrostella*, and *Trichaster*, but five of them were only tentatively included, and *Trichaster* today is regarded as a synonym of *Geastrum*.

Radiigera Zeller is one of the genera traditionally considered closely related to *Geastrum*, but is morphologically distinct on account of the non-splitting exoperidium and the poorly developed endoperidium (Sunhede, 1989; Domínguez de Toledo & Castellano, 1996). *Radiigera* contains three Nearctic and one Palearctic (Domínguez de Toledo & Castellano, 1996), plus one paleotropical species (Orihara & al., 2008). Previous phylogenetic analyses in the order Geastrales have shown *Radiigera* specimens to be nested in *Geastrum* (Hosaka & al., 2007; Hosaka & Castellano, 2008; Da Silva & al., 2013), but none of these studies was specifically focused on the relationship between the two genera. However, Jeppson & al. (2013) have recently considered *Radiigera* as a heterotypic synonym of *Geastrum*, and have combined the species previously included in *Radiigera* into *Geastrum*. *Myriostoma* is almost invariably treated as a well-defined genus, and its morphological boundaries are clearly established (Sunhede, 1989). However, its phylogenetic separation from *Geastrum* is not fully supported (Hosaka & Castellano, 2008; Kasuya & al., 2012; Da Silva & al., 2013; Jeppson & al., 2013).

The earthstars have attracted the attention of numerous mycologists, and Vittadini (1842), Hollós (1903), Staněk (1958), Dissing & Lange (1961), Dörfelt (1985), and Sunhede (1989) published rather comprehensive studies focused on the European species. Dissing & Lange (1962) studied the Congolese species, and recently Zhou & al. (2006) presented a conspectus of Chinese taxa. In the New World, among many others, Coker (1924), Long & Stouffer (1948), and Pérez-Silva & al. (1999) treated the North American earthstars, and Soto & Wright (2000) revised some Argentinian species. Cunningham (1944) monographed the Oceanian taxa, and Hemmes & Desjardin

(2011) studied the species of the Hawaiian Islands. De Toni (1887a, b), Lloyd (1902), and Ponce de León (1968) prepared world-wide revisions.

Despite this high number of studies of *Geastrum*, only few infrageneric subdivisions of the genus have been proposed, all of them based exclusively on morphological characters of the fruit bodies or on growth habit (Table 1). Some authors gave different weight to certain traits, resulting in rather different infrageneric units. First, Fries (1829) considered peristome structure the most important feature for subdividing the genus into four groups, *Myriostoma*, *Plecostoma*, *Geastrum*, and *Actinodermium*. Spegazzini (1881) considered both the peristome and the presence or absence of a stalk connecting the endoperidium and the exoperidium, and Morgan (1884), De Toni (1887a), Morgan (1889), and Hollós (1903) added the shape of the exoperidium for making their classifications. For Lloyd (1902), the hygrometric behaviour of the exoperidium and its consistence (harder when hygrometric) were the most important characters for dividing *Geastrum* into two sections, *Rigidae* and *Non-rigidae*. Later, Staněk (1958) started to use the ability of the mycelial layer to encrust debris, which was incorrectly interpreted as the mycelium remaining attached to the base of the young fruit body or enclosing it completely, as the most important character to divide *Geastrum* into two sections, *Basimyceliata* and *Perimyceliata*, and used the peristome type to divide each section into two subsections, *Laevistomata* and *Plicostoma* in sect. *Basimyceliata*, and *Glabrostomata* and *Sulcostomata* in sect. *Perimyceliata*. Dissing & Lange (1962), Ponce de León (1968), and Dörfelt (1985) considered most of the morphological traits listed above plus habit or growth form, the presence or absence of a myceliar subiculum, and whether the endoperidium remains attached to the pseudoparenchymatous layer (or, as they described it, whether the endoperidium is “fugacious” or disintegrates at maturity). Sunhede (1989), who undertook the most comprehensive study of the genus until now and focused on the European species, did not propose a new infrageneric classification, but accepted the sections proposed by Staněk (1958). A consensus about the infrageneric subdivision of *Geastrum* has not yet been reached, resulting in taxonomic treatments (e.g., Zhou & al., 2006; Hemmes & Desjardin, 2011; Jeppson, 2013; Jeppson & al., 2013) that avoid any subdivision or even consideration of species groups.

Few previous molecular studies have focused on Geastraceae. Douanla-Meli & al. (2005) and Da Silva & al. (2013) analysed nrLSU or a combination of nrLSU and *atp6*, respectively, to justify the recognition of new species. Kasuya & al. (2012) presented a study using ITS, nrLSU, and *atp6* to demonstrate the polyphyly of previously known species. Their results

Fig. 1. Morphological variation of basidiomata in Geastraceae (field photographs). **A**, *Myriostoma coliforme*; **B**, *Geastrum fimbriatum* (sect. *Fimbriata*); **C**, *Geastrum fornicatum* (sect. *Fornicata*); **D**, *Geastrum coronatum* (subsect. *Geastrum*); **E**, *Geastrum parvistriatum* (subsect. *Sulcostomata*); **F**, *Geastrum* aff. *arenarium* (subsect. *Arenaria*); **G**, *Geastrum campestre* (sect. *Campestris*); **H**, *Geastrum floriforme* (sect. *Papillata*); **I**, *Geastrum melanocephalum* (sect. *Trichaster*); **J**, *Geastrum elegans* (sect. *Elegantia*); **K**, *Geastrum rufescens* (sect. *Exareolata*); **L**, *Geastrum velutinum* agg. (subsect. *Velutina*); **M**, *Geastrum violaceum* (subsect. *Plicostomata*); **N**, *Geastrum corollinum* (subsect. *Marginata*); **O**, *Geastrum lageniforme* (subsect. *Lageniformia*). — Photos by J.C. Zamora.



Table 1. Major infrageneric classifications of *Geastrum* and characters used.

Reference	Fries (1829)	Spegazzini (1881)	Morgan (1884)	De Toni (1887a)	Morgan (1889)	Lloyd (1902)	Hollós (1903)
Characters	Peristome	Peristome and stalk	Peristome, stalk and exoperidial shape	Peristome, stalk and exoperidial shape	Peristome, stalk and exoperidial shape	Exoperidial hygrometry and consistence	Peristome, stalk, and exoperidial shape
Taxonomic units	<i>Myriostoma</i> <i>Plecostoma</i> <i>Geastrum</i> <i>Actinodermium</i>	<i>Eugeaster</i> <i>Plecostoma</i> <i>Odontostoma</i> <i>Aplostoma</i> <i>Schizogaster</i> <i>Heterogaster</i>	<i>Fornicati</i> <i>Striati</i> <i>Limbati</i> <i>Fimbriati</i> <i>Dentati</i> <i>Hygrometrici</i>	Sect. <i>Columnati</i> Sect. <i>Fornicati</i> Sect. <i>Cupulati</i> Sect. <i>Striati</i> Sect. <i>Fimbriati</i> Sect. <i>Papillati</i> Sect. <i>Exareolati</i>	<i>Depelliti</i> <i>Pedicellati</i> <i>Sessiles</i> <i>Pelliculosi</i> <i>Saccati</i>	Sect. <i>Rigidae</i> Sect. <i>Non-rigidae</i>	<i>Striati</i> <i>Pedicellati</i> <i>Sessiles</i> <i>Fimbriati</i> <i>Fornicati</i> <i>Sessiles</i> <i>Pedicellati</i>

showed that a single morphological feature (the pseudoparenchymatous collar characteristic of so-called *G. triplex*) is homoplastic instead of representing a synapomorphy for a species or a species group. However, a comprehensive morphological study was not performed and it remained unknown whether the different taxa are distinguishable morphologically. Finally, Jeppson & al. (2013) used ITS, nrLSU and *tef-1a* to elaborate a rather detailed phylogeny of the European taxa. These studies recovered *Geastrum* as monophyletic, but the distinctness of *Myriostoma* was not supported in all analyses (Kasuya & al., 2012; Da Silva & al., 2013, Jeppson & al., 2013).

The two aims of the present study are: (1) to establish morphological and phylogenetic boundaries among *Geastrum*, *Myriostoma*, and *Radiigera*; and (2) to propose a revised infrageneric classification of *Geastrum* based on molecular phylogenetic data and supported by morphological and chemical traits.

■ MATERIALS AND METHODS

Sampling. — A total of 139 samples, corresponding to roughly 80 species, were selected from about 3000 revised collections, trying to cover the entire range of morphological variation (Fig. 1). Species recognition was mainly based on Sunhede's (1989) taxonomic treatment, but other works were followed for several non-European species (e.g., Lloyd, 1902; Dissing & Lange, 1962; Ponce de León, 1968). Although discussion of the species delimitation in *Geastrum* is not part of the scope of the present study, we mostly followed phylogenetic species recognition by genealogical concordance (Taylor & al., 2000). The following herbaria were studied: BAFC, COFC, K, LPS, MA-Fungi, MICH, NY, S, UFRN, UPS, and the private herbaria of R. Lebeuf, M.Á. Ribes, S. Sunhede, and J.C. Zamora. The genus *Schenella* (Schenellaceae Nann.-Bremek.) was selected as outgroup based on Hosaka & al. (2007). Geographic origin, voucher information, and GenBank accession numbers of the studied specimens are summarized in Appendix 1.

Morphological and macrochemical study. — Specimens were studied following the classical methods described by Sunhede (1989) and Calonge (1998). Terminology follows those works with minor exceptions.

Micromorphological studies were carried out using a light microscope and scanning electron microscope. For light microscopy, small pieces of different parts of the basidiomata and rhizomorphs were mounted in 5% KOH or Hoyer's medium and observed with a Reichert-Jung or a Jeulin light microscope at 400× or 1000× magnification. Measurements were all taken at 1000× magnification. For scanning electron microscopy, air-dried samples were mounted on a sample holder covered with double-sided adhesive tape, coated with pure gold in a Balzers SCD 004 sputter coater, and observed with a Hitachi S-3000N SEM.

Data concerning macrochemical spot tests with 1-naphthol, guaiac gum, and syringaldazine are taken from Zamora & al. (2013).

DNA isolation, primer design, PCR amplification, DNA purification, and sequencing. — DNA was extracted from <10 mg of mature glebal masses, either with E.Z.N.A. Fungal DNA Miniprep Kit (Omega Biotek, Doraville, Georgia, U.S.A.) or DNeasy Plant Mini Kit (Qiagen, Valencia, California, U.S.A.), following the manufacturers' instructions, with minor modifications. Basidiospores were disrupted in microcentrifuge tubes with a micropestle after suspension in the lysis buffer of the respective kit or using Tyssue Lyser II (Qiagen) and glass beads.

Four molecular markers were investigated, i.e., ITS (including 5.8S) and 28S (LSU) nrDNA, the largest subunit of RNA polymerase II (*rpb1*), and subunit 6 of ATP synthase (*atp6*). The complete ITS nrDNA region, which comprises ITS1 (internal transcribed spacer 1), the 5.8S rRNA, and the ITS2 (internal transcribed spacer 2), was amplified using the primer pair ITSIF (Gardes & Bruns, 1993)/ITS4 (White & al., 1990). For amplifying the LSU nrDNA region, primer combination LR0R and either LR5 or LR7 (Vilgalys & Hester, 1990) was used. The *rpb1* region was initially amplified with primer pair gRPB1-A (Stiller & Hall, 1997)/fRPB1-C (Matheny & al., 2002). However, due to low amplification success, two degenerated

Table 1. Continued.

Reference	Staněk (1958)	Dissing & Lange (1962)	Ponce de León (1968)	Dörfelt (1985)	Zamora & al. (present study)
Characters	Mycelial layer and peristome	Combination of morphological traits	Combination of morphological traits	Combination of morphological traits	Combination of morphological traits, phenoloxidas, and molecular data
Taxonomic units	Sect. <i>Perimyceliata</i> Subsect. <i>Glabrostomata</i> Stirps <i>Fimbriatum</i> Stirps <i>Floriforme</i> Stirps <i>Coronatum</i> Stirps <i>Quadrifidum</i> Subsect. <i>Sulcostomata</i> Stirps <i>Pectinatum</i> Stirps <i>Campestre</i> Sect. <i>Basimyceliata</i> Subsect. <i>Laevistomata</i> Stirps <i>Triplex</i> Stirps <i>Smardae</i> Stirps <i>Welwitschii</i> Subsect. <i>Plicostomata</i>	Emendation of Staněk's sect. <i>Basimyceliata</i> and subdivision as follows: Sect. <i>Basimyceliata</i> Subsect. <i>Epigaea</i> Stirps <i>subiculosum</i> Stirps <i>stipitatum</i> Stirps <i>Lloydii</i> Stirps <i>rhizophorum</i> Subsect. <i>Subhypogaea</i> Stirps <i>velutinum</i> Stirps <i>triplex</i> Stirps <i>Smardae</i> Stirps <i>Welwitschii</i> Subsect. <i>Trichaster</i>	Subg. <i>Geastrum</i> Sect. <i>Geastrum</i> Subsect. <i>Geastrum</i> Group <i>Coronatum</i> Group <i>Fimbriatum</i> Group <i>Floriforme</i> Group <i>Minimum</i> Group <i>Quadrifidum</i> Subsect. <i>Sulcostomata</i> Group <i>Pectinatum</i> Group <i>Campestre</i> Group <i>Lloydianum</i> Sect. <i>Basimyceliatum</i> Subsect. <i>Basimyceliatum</i> Group <i>Indicum</i> Group <i>Smardae</i> Subsect. <i>Marginatum</i> Group <i>Recolligens</i> Subg. <i>Myceliostroma</i> Sect. <i>Myceliostroma</i> Group <i>Schweinitzii</i> Sect. <i>Subepigaea</i> Group <i>Javanicum</i> Subg. <i>Trichaster</i> Sect. <i>Trichaster</i> Group <i>Melanocephalum</i> Sect. <i>Lignicola</i> Group <i>Xylogenum</i>	Subg. <i>Trichaster</i> Subg. <i>Geastrum</i> Subg. <i>Pectinata</i> Subg. <i>Myceliostroma</i>	Sect. <i>Campestris</i> Sect. <i>Corollina</i> Subsect. <i>Lageniformia</i> Subsect. <i>Marginata</i> Subsect. <i>Plicostomata</i> Sect. <i>Elegantia</i> Sect. <i>Exareolata</i> Sect. <i>Fimbriata</i> Sect. <i>Fornicata</i> Sect. <i>Geastrum</i> Subsect. <i>Arenaria</i> Subsect. <i>Geastrum</i> Subsect. <i>Quadrifida</i> Subsect. <i>Sulcostomata</i> Sect. <i>Hariotia</i> Sect. <i>Hieronymia</i> Sect. <i>Myceliostroma</i> Subsect. <i>Epigaea</i> Subsect. <i>Velutina</i> Sect. <i>Papillata</i> Sect. <i>Pseudolimbata</i> Sect. <i>Schmidelia</i> Sect. <i>Trichaster</i>

internal primers were designed: RPBIGEA-1F (5' GTCAAGT CAAAAAGATAYTGGA 3') and RPBIGEA-2r (5' TYACCT GYAARAGYTGTCTC 3'). Some amplification reactions were performed by nested PCR, using *rpb1* general primers in a first PCR, and 1 µl of this PCR product as DNA template for a second PCR using the newly designed primers. Finally, for amplifying *atp6*, primer pairs *atp6-1/atp6-2* or *atp6-3/atp6-2* (Kretzer & Bruns, 1999) were used. For the few samples that could not be amplified using the cited primers two new internal primers were designed to amplify *atp6* in two fragments. The first fragment was amplified with *atp6-3* and the new primer ATP6GEA-ir (5' GTHACACCTATAAATAAGTAHHHC 3'), and the second fragment was amplified using the new primer ATP6GEA-iF (5' GDDTTACTATHTTTATAGGTGTDAC 3') and *atp6-2*.

PCRs were done in a final volume of 25 µl either with Ready-To-Go PCR Beads (Amersham Biosciences, Little Chalfont, England), with BIOTAQ (Ecogen, Barcelona, Spain), or with HotStarTaq MasterMix (Qiagen), following the manufacturers' instructions and using 1 µl of each primer (10 µM), in an MJ Research thermocycler or in an Eppendorf thermocycler.

The cycling parameters for ITS amplification have been described in Martin & Winka (2000). Cycling parameters for LSU were an initial step at 94°C for 2–15 minutes (depending on the polymerase), followed by 35 cycles of denaturation at 94°C for 30 s, annealing at 52°C for 30 s and extension at 72°C for 1.5 min, with a final extension at 72°C for 10 min. Cycling parameters when using general primers of *rpb1* were an initial step at 94°C for 2–15 min (depending on the polymerase), followed by 35 cycles of denaturation at 94°C for 30 s, annealing at 54°C for 30 s and extension at 72°C for 1.5 min, with a final extension at 72°C for 10 min. When using the specific primers the annealing temperature was increased to 58°C, with the remaining parameters unchanged. Finally, cycling parameters for *atp6* were those described by Kretzer & Bruns (1999), also when amplifying the marker with the new internal primers.

PCR products were checked on 2% agarose DI low EEO (CONDA, Pronadisa) gels and purified using the QIAquick Gel Extraction Kit (Qiagen), QIAquick PCR Purification Kit (Qiagen), or ExoSap-IT (VWR, Barcelona, Spain), following the manufacturers' instructions. Sequencing was carried out by MacroGen (Amsterdam, The Netherlands) or Secugen (Madrid,

Spain) services, or using the Big Dye Terminator Cycle Sequencing Kit (Applied Biosystems, Norwalk, Connecticut, U.S.A.), following the manufacturer's instructions.

Sequence alignment and phylogenetic analyses. — DNA sequence edition and consensus assembly were performed using Sequencher v.4.1.4 (Gene Codes, Ann Arbor, Michigan, U.S.A.). Sequences were primarily aligned through the MAFFT v.7 (Kato & al., 2002) on-line server, using the FFT-NS-i strategy, and manually corrected with BioEdit v.7.0 (Hall, 1999), with the exception of the *atp6* dataset that was directly aligned with BioEdit when transcribed to a protein. The variable ITS1 and ITS2 regions were excluded from the analyses due to difficulties to align them without ambiguities. An ambiguously aligned region of *atp6* of 69 bp was directly removed, while the few ambiguously aligned regions of the nrLSU and *rpb1* alignments were removed with Gblocks v.0.91b (Castresana, 2000), keeping the default settings but allowing all gap positions when not ambiguous. From the nrLSU and *rpb1* alignments, 42 and 112 positions respectively were removed. The remaining indels in each marker were coded using the simple indel coding method of Simmons & Ochoterena (2000), implemented in FastGap v.1.2 (Borchsenius, 2007), in a separate binary data subset. Data matrices are available in TreeBase (<http://www.treebase.org>; TB2:S15864).

GenBank sequences from previous studies were not included in the analyses for three main reasons. First, we do not know the morphology of the specimens from which those GenBank sequences were obtained, and our work is based on morphological features that are normally overlooked in most works. Second, *rpb1* is the marker that provided best phylogenetic resolution, but no sequences of this marker were found in GenBank. Third, we detected many misidentifications when comparing our sequences with GenBank sequences, so that names present in that database are not reliable.

Phylogenetic analyses were performed using three different strategies, i.e., maximum parsimony (MP), maximum likelihood (ML), and Bayesian inference (BI). Dataset congruence was assessed analysing each dataset with the “fast” stepwise-addition parsimony bootstrapping procedure implemented in PAUP* v.4.0b10 (Swofford, 2003), performing 1000 non-parametric bootstrap replicates (Felsenstein, 1985). Conflict between datasets was considered to exist when a significantly supported clade (bootstrap support $\geq 70\%$) from one marker was contradicted by another significantly supported clade from a different marker (Hillis & Bull, 1993). When no conflicts were detected, the different datasets were combined.

MP analysis was conducted with PAUP* v.4.0b10 (Swofford, 2003), using PaupUp v.1.0.3.1 graphical interface (Calendini & Martin, 2005), performing heuristic searches with “MulTrees” in effect, TBR as branch-swapping algorithm, MaxTrees = 1000, and branches collapsing when maximum branch length is zero. All characters were equally weighted and coded indels were treated as binary data. Non-parametric bootstrap searches (Felsenstein, 1985) were used to calculate branch support (MPbs), performing 1000 replicates with the full-heuristic option and the same parameters used in the heuristic search. If more than one MP tree was obtained, the likelihood-based

Shimodaira-Hasegawa topology test (Shimodaira & Hasegawa, 1999) was used to determine whether any of the MP trees was significantly better than others, using RELL approximation and 1000 bootstrap replicates.

The ML analysis was run in GARLI v.2.0 (Zwickl, 2006), partitioning the different markers as separate subsets, using the GTR+I+ Γ model for each DNA data subset, and the symmetric one-rate Mk model for the binary data subset (Lewis, 2001). The analysis was repeated twice starting from random trees. As in the MP analysis, 1000 non-parametric bootstrap replicates were performed to assess branch support (MLbs).

The Bayesian analysis was done using the Metropolis Coupled Markov Chain Monte Carlo (MC³) strategy implemented in MrBayes v.3.2 (Ronquist & al., 2012), through the University of Oslo Biportal on-line service (Kumar & al., 2009, <http://www.biportal.uni.no>). The different genes were partitioned as separate subsets, with topology linked across subsets, but separate model parameter values for each. Nucleotide substitution models were selected with jModelTest v.2.1 (Darriba & al., 2012), using the Akaike information criterion (AIC) and the Bayesian information criterion (BIC). When both criteria did not choose the same model, the model with the higher average likelihood score was selected. The F81 model for restriction sites was used for the binary data subset (coded gaps). Four parallel runs were executed, each one starting with a random tree, with six chains, five of which were incrementally heated with a temperature of 0.1 (we detected mixing problems with the default 0.2 value), and length preset to 10^7 generations, sampling every 100th tree. Runs were automatically stopped when the average standard deviation across runs dropped below 0.005. Convergence of the MC³ was additionally assessed with Tracer v.1.4 (Rambaut & Drummond, 2007) to check that values of effective sample size for each parameter were ≥ 200 . The first 50% of the analysis was discarded as burn-in. The 50% majority-rule tree with branch lengths and posterior probabilities (pp), indicating branch support, was calculated from the post-burn-in trees.

With the default priors serious convergence problems were detected. After repeating the analysis three times, the four simultaneous runs converged properly only once (in the other two analyses one run got trapped sampling trees with much higher branch lengths than in the other three runs). However, even in the analysis where the four runs converged, branch length estimates of the Bayesian analysis were more than 1.5 times higher than in the ML analysis, and the tree length (TL) value of the ML tree was completely outside the TL value interval of the Bayesian sampled trees. Since we aimed at more accurate estimates, the analysis was repeated using a correction for the λ parameter that controls the exponential prior on branch lengths (Brown & al., 2010; Marshall, 2010). Following Meseguer & al. (2013), but considering that our problem with branch lengths was not severe (not orders or magnitude), the default value $\lambda = 10$ was increased to $\lambda = 50$, with the command “prset brlenspr = unconstrained:exp(50)”. Using this prior, convergence of the analysis was improved and accelerated, values of branch length estimates were much closer to that of the ML analysis, the harmonic mean

of the likelihood was increased, and the posterior probabilities of some clades were increased, too. This last analysis with “corrected” branch lengths will be presented and discussed.

Interpretation of branch support values follows the criteria and scale defined in Lutzoni & al. (2004). Phylogenetic trees were drawn with FigTree v.1.3 (Rambaut, 2007).

■ RESULTS

A total of 529 new sequences were generated for this study (127 5.8S, 135 nrLSU, 135 *rpbl*, and 132 *atp6* sequences; see Appendix 1). Test for congruence after maximum parsimony bootstrapping revealed no significant conflicts among the different markers, and the four datasets were concatenated for further phylogenetic analyses.

The final matrix of the combined dataset had 3252 characters (155 in the 5.8S, 1002 in the nrLSU, 1152 in the *rpbl*, 671 in the *atp6*, and 272 in the coded indels datasets), 1714 of which were constant, 1303 parsimony-informative, and 235 parsimony-uninformative. Search for trees with MP retained 18 best trees, all with a rather similar topology, and with a length of 8090 steps. The Shimodaira-Hasegawa test did not reveal any tree as significantly better than others. Parsimony tree scores were identical for all trees: consistency index = 0.287, retention index = 0.710, rescaled consistency index = 0.204, and homoplasy index = 0.713.

The phylogenetic reconstructions from the ML analysis gave a best tree with an overall topology similar to that of the MP trees concerning well-supported clades, with minor variations in part of the backbone and some unsupported groups. Therefore, only the ML tree is shown in Fig. 2, with both MPbs and MLbs values. The likelihood scores for the two runs after final optimization in GARLI were $\ln L_1 = -43,402.1766$ (best) and $\ln L_2 = -43,402.1774$, respectively. Both trees showed an identical topology.

According to the AIC (that yielded the higher average likelihood scores), the substitution models selected by jModel-Test for the Bayesian analysis were SYM+I+ Γ for 5.8S and GTR+I+ Γ for nrLSU, *rpbl*, and *atp6*.

The Bayesian MC³ analysis was automatically halted after 2,545,000 generations, when the deviation of split frequencies across runs dropped below 0.005. Best likelihood states for the four runs were: $\ln L_1 = -43,422.71$, $\ln L_2 = -43,429.09$, $\ln L_3 = -43,440.68$, and $\ln L_4 = -43,453.25$. Potential scale reduction factor values for all model parameters were between 1.000 and 1.002. At this stage, 101,804 trees had been sampled, and the first 50% were discarded as burn-in. The 50% majority-rule consensus tree was calculated from the 50,904 post-burn-in trees. This consensus tree had an almost identical topology to that of the ML tree. It is therefore not shown but pp values are indicated on the branches of the ML tree (Fig. 2).

Geastraceae is recovered as monophyletic and strongly supported (MPbs = 100%, MLbs = 100%, pp = 1.0), including *Geastrum* and *Myriostoma*. *Geastrum* is recovered as a monophyletic genus and as distinct from *Myriostoma*, with high bootstrap support from the MP analysis (MPbs = 100%), but not

well-supported by the ML and Bayesian analyses (MLbs = 51%, pp = 0.88). Within *Geastrum*, 14 clades, which will be treated as different sections (see taxonomic part), were found, and all of them were strongly to moderately supported when the results from the three different phylogenetic analyses were combined. Branch support for all clades is summarized in Table 2. The sect. *Geastrum*, sect. *Myceliostroma*, and sect. *Corollina* clades are further divided into subclades (considered different subsections, see taxonomic part), as also indicated in Table 2.

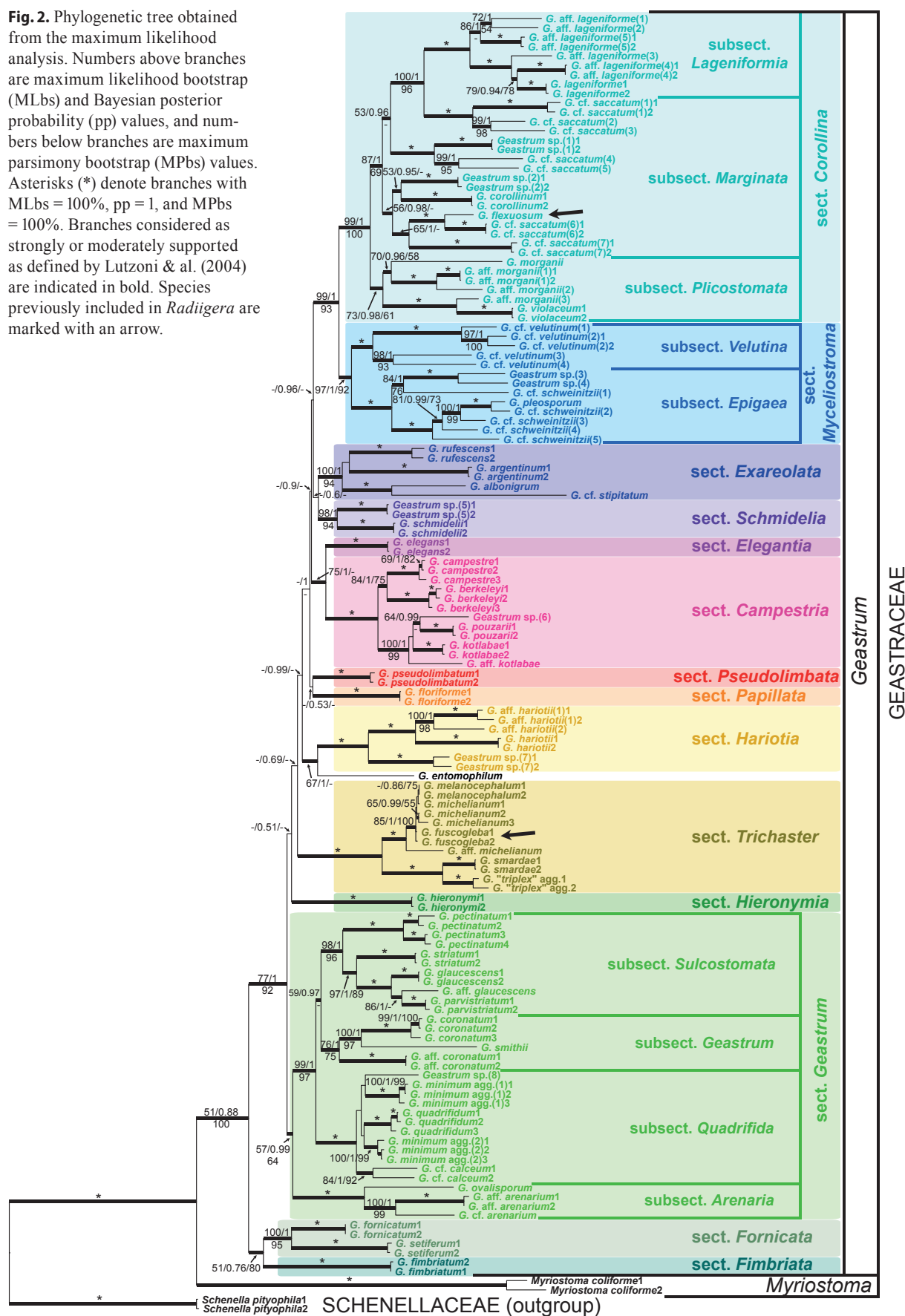
As noted above, tree topologies were almost identical between the ML and Bayesian trees, but somewhat different mainly concerning the backbone in the MP tree. However, these topological differences do not affect the proposed infrageneric subdivision, none is supported by bootstrapping, and some are also not supported in the Bayesian analysis (Fig. 2).

■ DISCUSSION

Towards a more natural classification of the earthstars: the consensus among morphological, chemical, and molecular phylogenetic data. — All previous infrageneric taxonomic treatments of *Geastrum* (Table 1) are highly incongruent with the phylogenetic results obtained here. Features traditionally used to define the higher infrageneric groups (e.g., ability of the mycelial layer to encrust debris or peristome structure) are present in different clades across the tree, or even mixed within individual clades. However, paying more attention to whole set of morphological traits, some natural morphological units can be defined. Recently discovered characters such as macrochemical reactions detecting phenoloxidase activity, or rhizomorph features, and particularly calcium oxalate crystals (Zamora & al., 2013), have not been used previously to subdivide *Geastrum*. Endoperidial surface ornamentation (Fig. 3) has been used only to establish some “groups” or “stirps” (Staněk, 1958; Ponce de León, 1968). However, they are valuable data that support rather well most of the phylogenetic relationships observed here. Those characters together with most classical morphological characters (i.e., peristome or mycelial layer), allowed us to propose a more natural infrageneric classification.

We consider the system presented detailed enough to accommodate most described species in a section, and flexible enough to group sections into subgenera in the future, or to add new infrageneric groups. Proposing subgenera is discouraged because phylogenetic relationships among most of the sections remain obscure (Fig. 2). However, there are two exceptions in which subgenera could be described with some confidence. The first is the clade formed by sect. *Fimbriata* and sect. *Fornicata*, which is well-supported in the MP analysis (MPbs = 80%) but not in the ML or Bayesian analyses (MLbs = 51%, pp = 0.76). It is sister to the rest of *Geastrum* and likely represents a good subgenus, mainly characterized by the endoperidial surface covered with simple, protruding hyphae, double-layered mycelial layer encrusted with debris and easily separable from the fibrous layer, pseudoparenchymatous layer not reacting with syringaldazine, and bipyramidal crystals on rhizomorphs (Zamora & al., 2013). Section *Myceliostroma* and

Fig. 2. Phylogenetic tree obtained from the maximum likelihood analysis. Numbers above branches are maximum likelihood bootstrap (MLbs) and Bayesian posterior probability (pp) values, and numbers below branches are maximum parsimony bootstrap (MPbs) values. Asterisks (*) denote branches with MLbs = 100%, pp = 1, and MPbs = 100%. Branches considered as strongly or moderately supported as defined by Lutzoni & al. (2004) are indicated in bold. Species previously included in *Radiigera* are marked with an arrow.



sect. *Corollina* form the other well-defined higher-level clade, with high statistical support (MPbs = 93%, MLbs = 99%, pp = 1.0). This clade could also be treated as a well-characterized subgenus, which groups all species with non-bipyramidal crystals as the main type of rhizomorph crystals, with the exception of *G. coronatum* that has horn-like crystals and does not belong to this group. Other synapomorphies are the saccate exoperidium and the sessile endoperidial body. In case a subgenus name would be used, the name subg. *Myceliostroma* Henn., with *G. juruense* Henn. as type, would be available (Hennings, 1904).

Morphological and chemical characteristics defining the proposed infrageneric taxa, as well as nomenclatural and taxonomic considerations are provided for each group in the taxonomic part.

***Geastrum*, *Myriostoma*, and *Radiigera* boundaries reassessed.** — As noted previously, the phylogenetic position of *Myriostoma* is well-supported in the MP analysis (MPbs = 100%), but poorly supported in the ML and Bayesian analyses (MLbs = 51%, pp = 0.88; Fig. 2). Our results are rather consistent with those of Kasuya & al. (2012) and Da Silva & al. (2013), who used *Sclerogaster* R.Hesse (Sclerogastraceae Locq. ex P.M.Kirk) instead of *Schenella* for rooting the tree, while Jeppson & al. (2013), who used *Sphaerobolus*, *Sclerogaster* and *Schenella*, indicated no MPbs support at all on the branch that separated *Geastrum* and *Myriostoma*. However, they did not show any results of ML analyses, and obtained either no (Kasuya & al., 2012; Jeppson & al., 2013) or very low (Da Silva & al., 2013) support for the branch that separated *Geastrum* and *Myriostoma* in the Bayesian analysis. It is known that phylogenetic accuracy and branch support may be altered due to several factors, such as Bayesian prior selection (Brown & al., 2010; Marshall, 2010), taxon sampling (Zwickl & Hillis, 2002), and number of markers used (Soltis & al., 2011). Therefore, the differences in support values in the backbone of the tree between Kasuya & al. (2012), Da Silva & al. (2013), Jeppson & al. (2013), and our analyses might be explained by our prior selection in the Bayesian analysis, the included taxa, and the addition of *rpbl* data. Incorporating additional markers might help to improve the resolution of this part of the tree. Our data suggest that *Myriostoma* can be considered a phylogenetic lineage different from *Geastrum*, and there is no molecular evidence to unite the two genera. *Myriostoma* and *Geastrum* can also be distinguished morphologically. *Myriostoma* basidiomata have several thin columellas intruding in the glebal mass, several stalks connecting the endoperidium and the exoperidium and, usually, multiple stomata in the endoperidium, with poorly developed peristomes; in addition, basidiospores are “alate”, ornamented with prominent, sinuous crests. In contrast, *Geastrum* basidiomata have a single columella, one or no stalk connecting the endoperidium and the exoperidium, and a single stoma in the endoperidium (except in teratogenic fruitbodies), surrounded by a rather well developed peristome; finally, basidiospores are always verrucose to pilate (Ponce de León, 1968; Sunhede, 1989; Calonge, 1998).

Concerning *Radiigera*, the two species included in the analysis are clearly nested in *Geastrum* as in previous studies

Table 2. Support values obtained in the MP, ML and Bayesian analyses of clades considered sections or subsections in the present study. Strength of support as defined by Lutzoni & al. (2004) is included. No support values are given for the paraphyletic subsect. *Marginata* (see taxonomic part). Taxa are numbered as in the taxonomic part.

Taxon	MPbs (%)	MLbs (%)	pp	Strength of support
(1) Sect. <i>Campestris</i>	100	100	1	Strong
(2) Sect. <i>Corollina</i>	100	99	1	Strong
(2a) Subsect. <i>Lageniformia</i>	100	100	1	Strong
(2b) Subsect. <i>Marginata</i>	—	—	—	—
(2c) Subsect. <i>Plicostomata</i>	61	73	0.98	Strong
(3) Sect. <i>Elegantia</i>	100	100	1	Strong
(4) Sect. <i>Exareolata</i>	94	100	1	Strong
(5) Sect. <i>Fimbriata</i>	100	100	1	Strong
(6) Sect. <i>Fornicata</i>	95	100	1	Strong
(7) Sect. <i>Geastrum</i>	64	57	0.99	Moderate
(7a) Subsect. <i>Arenaria</i>	100	100	1	Strong
(7b) Subsect. <i>Geastrum</i>	75	76	1	Strong
(7c) Subsect. <i>Quadrifida</i>	100	100	1	Strong
(7d) Subsect. <i>Sulcostomata</i>	96	98	1	Strong
(8) Sect. <i>Hariotia</i>	100	100	1	Strong
(9) Sect. <i>Hieronymia</i>	100	100	1	Strong
(10) Sect. <i>Myceliostroma</i>	92	97	1	Strong
(10a) Subsect. <i>Epigaea</i>	100	100	1	Strong
(10b) Subsect. <i>Velutina</i>	100	100	1	Strong
(11) Sect. <i>Papillata</i>	100	100	1	Strong
(12) Sect. <i>Pseudolimbata</i>	100	100	1	Strong
(13) Sect. <i>Schmidelia</i>	94	98	1	Strong
(14) Sect. <i>Trichaster</i>	100	100	1	Strong

(Hosaka & al., 2007; Hosaka & Castellano, 2008; Da Silva & al., 2013; Jeppson & al., 2013) (Fig. 2). Since one of the included species is the type of the genus (*R. fuscogleba*), *Radiigera* is here confirmed as a synonym of *Geastrum*, a conclusion previously noted by Jeppson & al. (2013) based on molecular data from GenBank sequences. This conclusion is also supported by the morphological characters used here to define different infrageneric units (see at the end of the taxonomic part “Comments on recent combinations of *Radiigera* in *Geastrum* from Jeppson & al. (2013)”). Considering that *Radiigera* basidiomata are not significantly different from unopened *Geastrum* basidiomata, this placement is not surprising, since sequestrate fungi seem to have evolved independently many times in different groups of fungi (Hibbet, 2007) such as Cortinariaceae (Peintner & al., 2001), Boletales (Binder & Hibbet, 2006), Russulales (Calonge & Martín, 2003; Eberhardt & Verbeken, 2004), and Phallomycetidae (Hosaka & al., 2007). What is more, *Radiigera* species do not form a clade but are placed in different sections (Fig. 2). Thus, the *Radiigera* morphology has been originated independently more than once in the evolution of *Geastrum*.

■ TAXONOMY

Geastrum Pers. in Neues Mag. Bot. 1: 85. 1794: Pers., Syn. Meth. Fung.: 131. 1801 (“*Geaster*” P. Micheli, Nov. Pl. Gen.: 220. 1729 ex Fries, Syst. Mycol. 3: 8. 1829 [see Art. 61.5 Ex. 3 and Demoulin, 1984]) $\equiv$ *Plecostoma* Desv., J. Bot. 2: 99. 1809 – **Type (designated here for *Plecostoma*):** *Geastrum coronatum* Pers. ($\equiv$ *Plecostoma coronatum* (Pers.) Desv. $\equiv$ *Geastrum multifidum* Pers.).
 $\equiv$ *Coilomyces* Berk. & M.A. Curtis in J. Acad. Nat. Sci. Philadelphia, ser. 2, 2: 279. 1853 – Type: *Coilomyces schweinitzii* Berk. & M.A. Curtis ($\equiv$ *Geastrum schweinitzii* (Berk. & M.A. Curtis) Zeller).

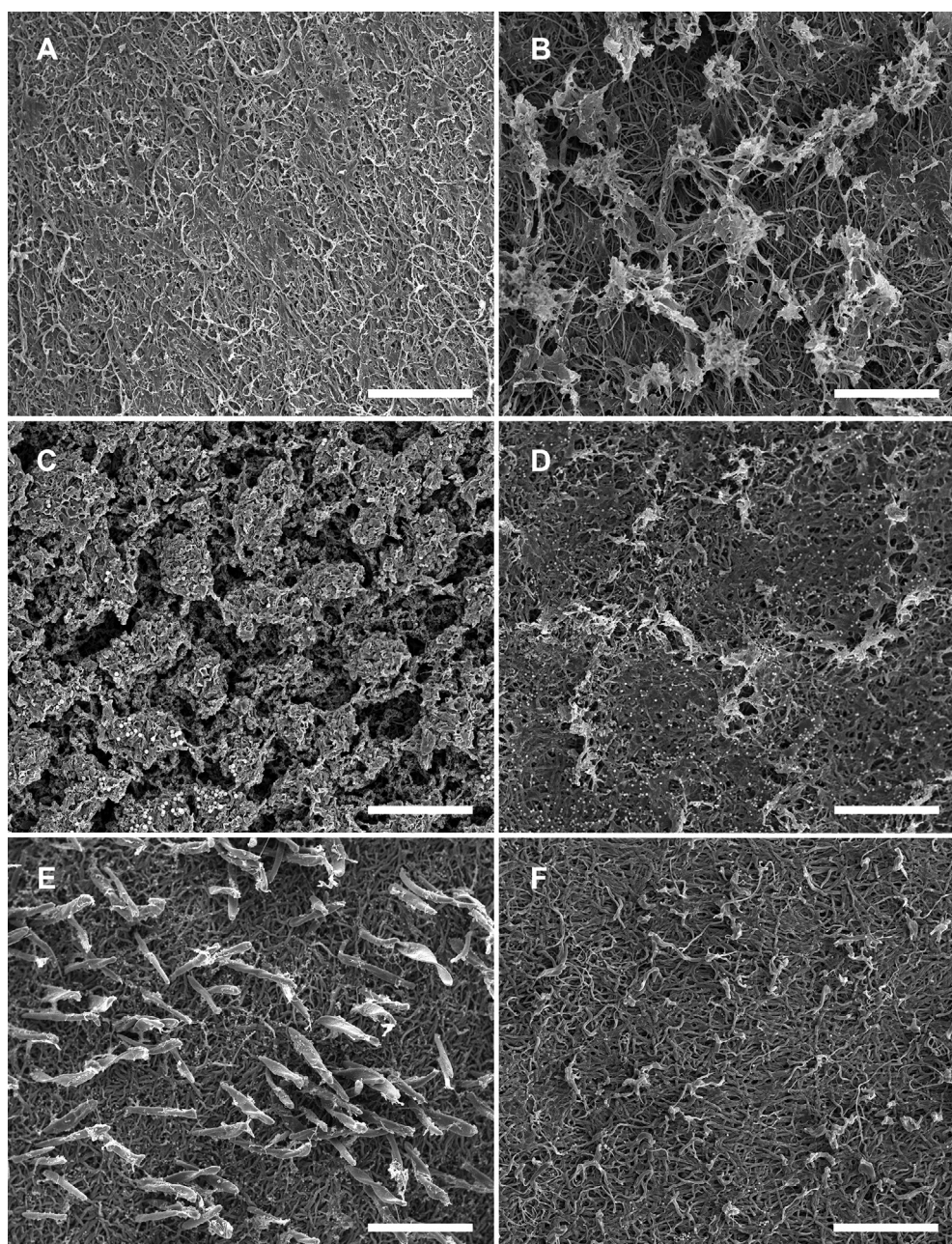
$\equiv$ *Cycloderma* Klotzsch in Linnaea 7: 203. 1832 – Type: *Cycloderma indicum* Klotzsch ($\equiv$ *Geastrum indicum* (Klotzsch) Rauschert).

$\equiv$ *Geasteroides* Battarra, Fungi Arimin.: 74. 1755, non Long in Mycologia 9: 217. 1917, nom. illeg. (Art. 53.1) – Type: not designated.

$\equiv$ *Radiigera* Zeller in Mycologia 36: 628. 1944 – Type: *Radiigera fuscogleba* Zeller ($\equiv$ *Geastrum fuscogleba* (Zeller) Jeppson & E. Larss.).

$\equiv$ *Trichaster* Czern. – Type: *Trichaster melanocephalus* Czern. (see *Geastrum* sect. *Trichaster* entry, p. 492).

Fig. 3. Endoperidial surfaces in six phylogenetic lineages of *Geastrum*. **A**, *Geastrum coronatum* (sect. *Geastrum*), almost smooth surface partially covered by collapsed generative hyphae from the mesoperidium; **B**, *Geastrum hieronymi* (sect. *Hieronymia*), strongly ornamented surface with irregular processes formed by bundles of hyphae; **C**, *Geastrum campestre* (sect. *Campestris*), strongly ornamented surface covered by more or less rounded warts; **D**, *Geastrum* aff. *hariotii* (sect. *Hariotia*), surface showing low elongated warts and ridges; **E**, *Geastrum setiferum* (sect. *Fornicata*), surface with very well-developed, erect, protruding hyphae (setulae); **F**, *Geastrum fimbriatum* (sect. *Fimbriata*), surface with small and bent protruding hyphae. — Scale bars = 150 μ m.



Nomenclatural notes on *Geastrum* typification

Although the typification of *Geastrum* and its orthographic variant *Geaster* had been resolved by Demoulin (1984), the changes made in Art. 10.2 of the *Melbourne Code* (McNeill & al., 2012) requires us to reconsider Clements & Shear's (1931) typification using *G. pectinatum* Pers. That is because Art. 10.2(b) now explicitly allows the selection of the type of a sanctioned generic name from both the protologue and the sanctioning treatment. Previously, the *Code* (e.g., McNeill & al., 2006) could be interpreted as requiring selection of the type of a sanctioned generic name from the protologue. When Demoulin (1984) typified *Geastrum*, the *Sydney Code* (Voss & al., 1983) was in effect, and Art. 7.17 of that *Code* stated that typification of sanctioned names may be effected in the light of anything associated with that name in the sanctioned work. However, Demoulin (1984) considered that Art. 10.2 overruled Art. 7.17 and typified *Geastrum* with *G. multifidum* ($\equiv$ *G. coronatum*), one of the species included in the protologue (Persoon, 1774). Now, since Clements & Shear's (1931) typification is in agreement with the current Art. 10.2(b), and it is the first one known so far, their choice should be followed (Art. 10.5). However, this type selection has not been subsequently followed. The next typification was made by Cunningham (1944), who selected *G. coronatum* Pers., the type that most authors have used through decades (e.g., Staněk, 1958; Ponce de León, 1968; Sunhede, 1989). In order to preserve the current usage of this typification, a proposal to conserve *Geastrum* with a conserved type (*G. coronatum*) has been prepared by the first author (Zamora, 2014). In accordance with Rec. 14A of the *ICN*, existing usage of the autonyms (*G.* sect. *Geastrum*, *G.* subsect. *Geastrum*) is maintained in the present study pending the General Committee's recommendation on the proposal because, otherwise, a new name for the subsection that includes *G. coronatum* would be needed. See also the Nomenclature Editor's footnote regarding the problematic status of *G.* subsect. *Sulcostomata* V.J.Staněk.

Key to the sections and subsections of *Geastrum*

1. Mycelial layer single-layered, exclusively of generative hyphae 2
1. Mycelial layer double-layered, internal layer of generative hyphae and external layer of skeletal hyphae 4
2. Exoperidium with well-developed stipe
..... sect. *Exareolata* (4) p.p.
2. Exoperidium without stipe 3
3. Pseudoparenchymatous layer reacting with syringaldazine; exoperidium saccate to arched or pseudofoeniculate; rhizomorphs with rose-like aggregates of bipyramidal crystals sect. *Trichaster* (14)
3. Pseudoparenchymatous layer not reacting with syringaldazine; exoperidium always saccate; rhizomorphs with mostly horn-like crystals
..... sect. *Corollina* (2) p.p. [subsect. *Lageniformia* (2a)]
4. Endoperidial surface rough due to the presence of warts, ridges or processes formed by several hyphae glued together 5

4. Endoperidial surface smooth or covered with individual protruding hyphae 9
5. Peristome fibrillose 6
5. Peristome sulcate 8
6. Exoperidium $\pm$ saccate; endoperidial body sessile
..... sect. *Papillata* (11)
6. Exoperidium arched; endoperidial body stalked 7
7. Exoperidium not hygrometric; pseudoparenchymatous layer with thin-walled (walls ≤ 1 μ m thick) hyphal cells; peristome indistinctly delimited and not thickened
..... sect. *Hieronymia* (9)
7. Exoperidium hygrometric; pseudoparenchymatous layer with thick-walled (walls ≥ 1.5 μ m thick) hyphal cells; peristome distinctly delimited and $\pm$ thickened
..... sect. *Pseudolimbata* (12)
8. Endoperidial surface with more or less rounded warts, sometimes coalescent; basidiospores 4.5–8.5 μ m diam., ornamentation 0.4–0.8 μ m high sect. *Campestris* (1)
8. Endoperidial surface with ridges or with more or less irregular and elongated warts; basidiospores 3.0–5.0 μ m diam., ornamentation 0.2–0.4 μ m high sect. *Hariotia* (8)
9. Exoperidium saccate, endoperidial body sessile 10
9. Exoperidium not saccate, endoperidial body distinctly stalked (stalk sometimes short) 16
10. Mesoperidium well developed, peristome truly sulcate ..
..... sect. *Elegantia* (3)
10. Mesoperidium absent or very poorly developed, peristome fibrillose to irregularly folded 11
11. Rhizomorphs covered mainly with bipyramidal crystals, often grouped in rose-like aggregates; peristome fibrillose and indistinctly delimited 12
11. Rhizomorphs covered mainly with acicular, horn-like or prism-shaped crystals; peristome fibrillose and distinctly delimited, or folded and indistinctly delimited 13
12. Endoperidium densely covered with protruding hyphae ("short hirsute"); basidiospores ≤ 4.5 μ m diam., light; pseudoparenchymatous layer not reacting with syringaldazine sect. *Fimbriata* (5)
12. Endoperidium glabrous or with sparse protruding hyphae; basidiospores ≥ 4.0 μ m diam., dark; pseudoparenchymatous layer reacting with syringaldazine
..... sect. *Exareolata* (4) p.p.
13. Species forming a more or less developed subiculum or a well-developed rhizomorph system, usually growing on rotten wood; pseudoparenchymatous layer reacting with syringaldazine; rhizomorphs with crystalliferous cystidioid elements with prism-shaped crystals
..... [sect. *Myceliostroma* (10)] 14
13. Species not forming a conspicuous subiculum, with or without abundant rhizomorphs, terrestrial; pseudoparenchymatous layer not reacting with syringaldazine; rhizomorphs without crystalliferous cystidioid cells but with long acicular to horn-like crystals
..... [sect. *Corollina* (2) p.p., excl. subsect. *Lageniformia*] 15
14. Lignicolous or, less frequently, humicolous species, forming a well-developed subiculum; basidiomata often caespitose, small to medium-sized (endoperidial body

- 3–15 mm diam.); rhizomorph cystidioid elements narrowly cylindrical, covered with coarse prism-shaped crystals ..
..... subsect. *Epigaea* (10a)
14. Humicolous species with a well-developed rhizomorph system; basidiomata not caespitose, medium-sized (endoperidial body 10–25 mm diam.); rhizomorph cystidioid elements more or less vesiculose, covered with narrow prism-shaped crystals subsect. *Velutina* (10b)
15. Peristome fibrillose, not folded, thickened and distinctly delimited subsect. *Marginata* (2b)
15. Peristome irregularly folded, not thickened and indistinctly delimited subsect. *Plicostomata* (2c)
16. Endoperidial surface with abundant protruding hyphae (“short-hirsute”) 17
16. Endoperidial surface glabrous 18
17. Pseudoparenchymatous layer reacting with syringaldazine; mycelial layer encrusting debris or not; exoperidium with pinkish or reddish tint; endoperidium with hyaline or light protruding hyphae; peristome always fibrillose
..... sect. *Exareolata* (4) p.p.
17. Pseudoparenchymatous layer not reacting with syringaldazine; mycelial layer always encrusting debris; exoperidium without pinkish or reddish tint; endoperidium with dark protruding hyphae; peristome fibrillose to finely sulcate
..... sect. *Fornicata* (6)
18. Basidia rounded, ellipsoid or clavate; mesoperidium very poorly developed, with scattered generative hyphae and some small crystals; peristome sulcate
..... sect. *Schmidelia* (13)
18. Basidia mostly narrowly cylindrical, ellipsoid or slightly lageniform; mesoperidium more developed (especially when peristome sulcate), formed by generative hyphae and/or crystals of various size; peristome fibrillose or sulcate...
..... [sect. *Geastrum* (7)] 19
19. Peristome always sulcate; mesoperidium a thick, often farinose, layer of pruina of generative hyphae and small crystals subsect. *Sulcostomata* (7d)
19. Peristome fibrillose to sulcate; mesoperidium different. 20
20. Basidiospores globose to ovoid, 3.0–5.0 µm diam., ornamentation 0.1–0.5 µm high subsect. *Arenaria* (7a)
20. Basidiospores globose to subglobose, 4.5–7.0 µm diam., ornamentation mostly 0.4–1.0 µm high 21
21. Basidiospores with processes not covered with warts or only with some irregular and inconstant warts; mesoperidium with crystalline matter more abundant than generative hyphae; peristome always fibrillose
..... subsect. *Quadrifida* (7c)
21. Basidiospores with processes densely covered by regular and small rounded warts; mesoperidium with generative hyphae more abundant than crystals; peristome fibrillose to sulcate subsect. *Geastrum* (7b)

Infrageneric classification

Species included in our molecular analyses are shown in bold in the “Included species” paragraph of each section or subsection. Species tentatively assigned to one of the proposed

infrageneric groups but not included in the molecular analyses are indicated non-bold. Only such species are considered when either a herbarium specimen was studied (*), or when the morphological traits were described with sufficient detail in the literature (name without symbol).

1. *Geastrum* sect. ***Campestris*** J.C.Zamora, **sect. nov.** – Type: *G. campestre* Morgan (Figs. 1G, 3C).
Mycobank MB 807734

Rhizomorphs with bipyramidal crystals, often forming rose-like aggregates, sometimes isolated. Exoperidium arched to somewhat saccate, hygrometric or not. Mycelial layer encrusting debris; double-layered, outer layer formed by skeletal hyphae and inner layer formed by generative hyphae. Endoperidial body sessile or stalked. Endoperidial surface more or less warty (sometimes very faintly). Mesoperidium moderately developed to almost absent. Peristome sulcate. Basidiospores 4.5–8.5 µm diam. and conspicuously ornamented (ornamentation 0.4–0.8 µm high), normally verrucose to irregularly baculate. Pseudoparenchymatous layer not reacting with gum guaiac.

Included species. – *Geastrum berkeleyi* Massee, *G. campestre* Morgan, *G. kotlabae* V.J.Staněk agg., *G. pouzarii* V.J. Staněk.

Discussion. – This section has previously been considered a natural group, corresponding exactly to Staněk’s (1958) “stirps *Campestre*” and Ponce de León’s (1968) “group *Campestre*”. Although morphologically similar to sect. *Hariotia*, basidiospores are smaller and inconspicuously ornamented in the latter section. In contrast, the hygrometric behaviour of the exoperidium, used by Ponce de León (1968) to separate “group *Campestre*” and “group *Lloydianum*” (equivalent to the present sect. *Hariotia*), seems to be unimportant.

2. *Geastrum* sect. ***Corollina*** J.C.Zamora, **sect. nov.** – Type: *G. corollinum* (Batsch) Hollós (Fig. 1M–O).
Mycobank MB 807735

Rhizomorphs with acicular or horn-like crystals, sometimes grouped to form arachnoid or candelabrum-like structures. Exoperidium mostly saccate and not hygrometric, rarely slightly arched and/or hygrometric. Mycelial layer normally not encrusting debris, rarely encrusted; single-layered and formed by generative hyphae, or double-layered, with an outer layer formed by skeletal hyphae, and an inner layer formed by generative hyphae. Endoperidial body sessile. Endoperidial surface glabrous to hirsute. Mesoperidium poorly developed to almost absent. Peristome fibrillose to irregularly folded, but not truly sulcate. Basidiospores 2.5–6.0 µm diam., mostly with a pilate-baculate ornamentation. Pseudoparenchymatous layer not reacting with syringaldazine.

Discussion. – Syringaldazine reaction and rhizomorph crystals are good differences to both sect. *Myceliostroma* and sect. *Trichaster* (see below). Since *G. saccatum* s.l. and *G. lageniforme* s.l. have almost invariably been considered similar to *G. triplex* s.l. (Lloyd, 1902; Staněk, 1958; Ponce de León, 1968; Dörfelt, 1985; Sunhede, 1989; Calonge, 1998), they have not been separated as their own section. We propose the new name sect. *Corollina* for this group. This large clade can be

divided into three morphologically homogenous subsections, two strictly monophyletic and one paraphyletic.

2a. *Geastrum* subsect. ***Lageniformia*** J.C.Zamora, **subsect. nov.** – Type: *G. lageniforme* Vittad. (Fig. 1O).
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Rhizomorphs with horn-like crystals often grouped in candelabrum-like structures. Mycelial layer single-layered, formed exclusively by generative hyphae. Peristome fibrillose, distinctly delimited and thickened.

Included species. – *Geastrum lageniforme* Vittad. agg.

Discussion. – Species in the *G. lageniforme* group have likely lost the external stratum of skeletal hyphae present in all other species of sect. *Corollina*. This was accompanied by the transformation of acicular crystals to thicker morphs, leading to the horn-like morphology characteristic of this subsection. Although subsect. *Lageniformia* and subsect. *Marginata* could be united, the recognition of subsect. *Lageniformia* seems valuable and helpful from a morphologically point of view. This decision is in agreement with the recognition of particular paraphyletic groups in classification, since paraphyly is inherent to certain evolutionary processes (Hörandl & Stuessy, 2010).

2b. *Geastrum* subsect. ***Marginata*** P.Ponce de León in Fieldiana, Bot. 31: 309. 1968 (“*Marginatum*”) – Type: *G. recolligens* (Sowerby) Desv. (= *G. corollinum* (Batsch) Hollós) (Fig. 1N).

Rhizomorphs with acicular crystals, rarely with some horn-like mixed. Mycelial layer double-layered, outer layer made up by skeletal hyphae and inner layer made up by generative hyphae. Peristome fibrillose, distinctly delimited and thickened.

Included species. – *Geastrum corollinum* (Batsch) Hollós, *G. flexuosum* (L.S.Domínguez & Castellano) Jeppson & E.Larss., *G. saccatum* Fr. agg.

Discussion. – This paraphyletic subsection is justified by the morphological homogeneity of all species included. The tree topology at this point is not fully resolved, and alternative topologies may appear with additional data.

2c. *Geastrum* subsect. ***Plicostomata*** V.J.Staněk in Pilát, Flora ČSR BI, Gasteromycetes: 782. 1958 – Type: *G. morganii* Lloyd (Fig. 1M).

Rhizomorphs with acicular crystals. Mycelial layer double-layered, outer layer made up by skeletal hyphae and inner layer made up by generative hyphae. Peristome irregularly folded, indistinctly to weakly distinctly delimited and not thickened.

Included species. – *Geastrum morganii* Lloyd agg., *G. violaceum* Rick.

Discussion. – The recently described *Geastrum reticulatum* (Hemmes & Desjardin, 2011) may belong to this subsection based on mycelial layer features and the not fibrillose peristome. However, the peristome of this species is truly and regularly sulcate, and rather well-delimited, deviating from the remaining species included in this group. Having no information about rhizomorph features and molecular data, we prefer to keep *G. reticulatum* unplaced.

3. *Geastrum* sect. ***Elegantia*** J.C.Zamora, **sect. nov.** – Type: *G. elegans* Vittad. (Fig. 1J).

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Rhizomorphs with bipyramidal crystals forming rose-like aggregates, rarely isolated. Exoperidium saccate to somewhat arched and not hygrometric. Mycelial layer encrusting debris; double-layered, with an outer layer formed by skeletal hyphae, and an inner layer formed by generative hyphae. Endoperidial body sessile. Endoperidial surface glabrous. Mesoperidium very well developed. Peristome sulcate. Basidiospores 4.5–7.0 µm diam., conspicuously ornamented (ornamentation 0.4–0.7 µm high), and verrucose to irregularly baculate. Pseudoparenchymatous layer not reactive with syringaldazine.

Included species. – *Geastrum elegans* Vittad.

Discussion. – *Geastrum elegans* is similar to subsect. *Sulcostomata* in its well-developed mesoperidium and sulcate peristome, but the morphology of basidia, shape of exoperidium, and absence of stalk justify exclusion from sect. *Geastrum* as suggested here. It appears more closely related to sect. *Campestris*, a relationship supported in the ML and Bayesian analyses, but not at all in the MP analysis (Fig. 2). Since the endoperidial surface and mesoperidium are rather different from those of sect. *Campestris*, we prefer to separate the species in a new section.

4. *Geastrum* sect. ***Exareolata*** De Toni in Rev. Mycol. (Toulouse) 9(34): 65. 1887 (“*Exareolati*”) – **Type (designated here):** *G. rufescens* Pers.: Pers. (Fig. 1K).

Rhizomorphs with bipyramidal crystals, isolated, forming rose-like aggregates, or covering poorly developed cystidioid elements, sometimes with some prism-shaped crystals intermixed. Exoperidium saccate to arched, not hygrometric. Mycelial layer encrusting debris or not; mostly double-layered, with an outer layer formed by skeletal hyphae, and an inner layer formed by generative hyphae, very rarely single-layered, then formed only by generative hyphae and basidiomata clearly stipitate. Endoperidial body sessile or stalked. Endoperidial surface glabrous or covered by thick-walled protruding hyphae. Mesoperidium almost absent. Peristome fibrillose, indistinctly delimited. Basidiospores 4.0–6.5 µm diam., baculate-pilate. Pseudoparenchymatous layer clearly reacting with syringaldazine.

Included species. – *Geastrum aculeatum* B.D.B.Silva & Baseia, *G. albonigrum* Calonge & M.Mata, *G. argentinum* Speg., *G. echinulatum* T.S.Cabral, B.D.B.Silva & Baseia, *G. litchiforme* Desjardin & Hemmes, *G. rufescens* Pers.: Pers., **G. scleroderma* Mont., *G. cf. stipitatum* Solms ex E.Fisch., **G. welwitschii* Mont.

Discussion. – This section is somewhat heterogeneous morphologically, uniting species with mycelial layer encrusting debris or not, exoperidium of various shapes (saccate, arched, fornicate), endoperidial body stalked or sessile, and endoperidial surface glabrous or not. However, the combination of mostly bipyramidal rhizomorph crystals, not delimited, fibrillose peristome, and positive reaction with syringaldazine on the pseudoparenchymatous layer characterize this group well.

5. *Geastrum* sect. ***Fimbriata*** De Toni in Rev. Mycol. (Toulouse) 9(34): 65. 1887 (“*Fimbriati*”) ≡ *Geastrum* subsect. *Glabrostomata* V.J.Staněk in Pilát, Flora ČSR BI, Gasteromycetes: 781. 1958 – Type: *G. fimbriatum* Fr. (Fig. 1B, 3F).

Rhizomorphs with bipyramidal crystals forming rose-like aggregates, rarely isolated. Exoperidium saccate, not hygrometric. Mycelial layer encrusting debris; double-layered, with an outer layer formed by skeletal hyphae, and an inner layer formed by generative hyphae. Endoperidial body sessile. Endoperidial surface hirsute, with isolated, thick-walled and light-coloured protruding hyphae. Mesoperidium absent to very poorly developed. Peristome fibrillose and indistinctly delimited. Basidiospores 3.0–4.5 µm diam., light-coloured, and irregularly baculate to pilate. Pseudoparenchymatous layer not reacting with syringaldazine.

Included species. – *Geastrum fimbriatum* Fr.

Discussion. – In addition to the characters listed, *G. fimbriatum* is the only species in the genus that often shows a positive reaction with 1-naphthol on the pseudoparenchymatous layer (Zamora & al., 2013). It was traditionally considered similar to *G. rufescens* Pers.: Pers. (Staněk, 1958; Ponce de León, 1968), which also has a “short” hirsute endoperidial surface but, according to our results, the two species are not closely related (Fig. 2). In *G. rufescens* basidiospores are darker, larger, and the reaction of the pseudoparenchymatous layer with syringaldazine is clearly positive, while in *G. fimbriatum* that reaction is negative.

6. *Geastrum* sect. ***Fornicata*** De Toni in Rev. Mycol. (Toulouse) 9(34): 64. 1887 (“*Fornicati*”) – Type: *G. fornicatum* (Huds.) Hook. (Fig. 1C, 3E).

Rhizomorphs with bipyramidal crystals forming rose-like aggregates, rarely isolated. Exoperidium fornicate to arched, not hygrometric. Mycelial layer encrusting debris; double-layered, with an outer layer formed by skeletal hyphae, and an inner layer formed by generative hyphae. Endoperidial body stalked. Endoperidial surface with isolated, thick-walled and dark-coloured protruding hyphae. Mesoperidium absent to very poorly developed. Peristome fibrillose to finely sulcate and indistinctly delimited. Basidiospores 4.0–5.0 µm diam., baculate to pilate. Pseudoparenchymatous layer not reacting with syringaldazine.

Included species. – *Geastrum dissimile* Bottomley, *G. fornicatum* (Huds.) Hook., *G. setiferum* Baseia.

Discussion. – The name sect. *Fornicata* was proposed by De Toni (1887a, as “*Fornicati*”) to include some fornicate species of the genus. However, our analyses show that the fornicate or pseudofornicate shape of the exoperidium is homoplastic for some species, such as *G. fornicatum* (sect. *Fornicata*), *G. quadrifidum* Pers. (sect. *Geastrum*), and *G. smardae* V.J.Staněk (sect. *Trichaster*), and that is not a synapomorphy for any section. The name can be used for this clade when excluding *G. radicans* and *G. welwitschii*, which have a mycelial layer not encrusting debris. Morgan (1884) previously used the name “*Fornicati*” for this group as well (although also including *G. triplex*), but without rank.

7. *Geastrum* Pers.: Pers. sect. ***Geastrum*** (Figs. 1D–F, 3A).

– “*Geastrum* sect. *Perimyceliata*” V.J.Staněk in Pilát, Flora ČSR BI, Gasteromycetes: 780. 1958, nom. inval. (Art. 22.2).

Rhizomorphs with bipyramidal crystals, isolated, forming rose-like aggregates, or covering poorly developed cystidioid elements; horn-like crystals sometimes present, but being very rarely the main type (*G. coronatum*). Exoperidium mostly arched, sometimes fornicate or almost saccate, rarely hygrometric. Mycelial layer encrusting debris; double-layered, with an outer layer formed by skeletal hyphae, and an inner layer formed by generative hyphae. Endoperidial body mostly stalked, sometimes almost sessile. Endoperidial surface glabrous. Mesoperidium moderately to very well developed. Peristome fibrillose to sulcate, or transitions between both types. Basidia mostly narrowly cylindrical, ellipsoid or slightly lageniform, slender. Basidiospores 3.0–7.0 µm diam., normally verrucose to irregularly baculate-pilate. Pseudoparenchymatous layer normally not reacting with syringaldazine, but sometimes reacting.

Discussion. – Although this clade seems heterogeneous from a morphological point of view, because it groups species with both sulcate and fibrillose peristomes, there are several synapomorphies such as slender basidia, smooth endoperidial surface, and well-developed mesoperidium. Since the peristome is one of the most important features traditionally used to subdivide *Geastrum*, no author has previously considered this group as natural. It contains the type of the genus, i.e., *G. coronatum*, so that its treatment as sect. *Geastrum* is appropriate. This clade has a well-defined phylogenetic structure and can easily be subdivided into four subsections.

7a. *Geastrum* subsect. ***Arenaria*** J.C.Zamora, **subsect. nov.** – Type: *G. arenarium* Lloyd (Fig. 1F).

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Mesoperidium moderately developed, formed mainly by crystalline matter. Peristome fibrillose. Basidiospores 3.0–5.0 µm diam. and often inconspicuously ornamented (ornamentation 0.1–0.5 µm high).

Included species. – *Geastrum arenarium* Lloyd agg., **G. leptospermum* G.F.Atk. & Coker, *G. ovalisporum* Calonge & Mor.-Arr.

Discussion. – Basidiospore features are the best character to define this subsection, which is rather similar to subsect. *Quadrifida* in all other morphological characters.

7b. *Geastrum* Pers.: Pers. subsect. ***Geastrum*** (Figs. 1D, 3A).

Mesoperidium moderately developed, formed mainly by hyphae strongly glued together, with or without some big crystals. Peristome fibrillose to sulcate, or transitions between both types. Basidiospores 4.5–6.5 µm diam., with a complex ornamentation (big processes covered by small warts).

Included species. – *Geastrum coronatum* Pers. agg., *G. smithii* Lloyd.

Discussion. – The rather complex and unique basidiospore ornamentation, typically formed by big processes that are minutely but clearly warted, is the best feature that characterizes this subsection.

7c. *Geastrum* subsect. *Quadrifida* J.C.Zamora, **subsect. nov.**
– Type: *G. quadrifidum* Pers.: Pers.

MycoBank MB 807739

Mesoperidium moderately developed, formed mainly by crystalline matter. Peristome fibrillose. Basidiospores 4.5–7.0 µm diam., with processes not clearly covered by small warts.

Included species. – *Geastrum calceum* Lloyd agg., *G. minimum* Schwein. agg., *G. quadrifidum* Pers.: Pers.

Discussion. – Subsection *Quadrifida* is morphologically similar to some species of subsect. *Arenaria* and subsect. *Geastrum*, differing mainly in basidiospore and mesoperidial features.

7d. *Geastrum* subsect. *Sulcostomata* V.J.Staněk in Pilát, Flora ČSR B1, Gasteromycetes: 781. 1958 ≡ *Geastrum* subg. *Pectinata* Dörfelt & Müller-Uri in Dörfelt, Die Erdsterne: Geastraceae und Astraeaceae: 17. 1985 – Type: *G. pectinatum* Pers. (Fig. 1E).¹

= *Geastrum* sect. *Striata* De Toni in Rev. Mycol. (Toulouse) 9(34): 64. 1887 (“*Striati*”) – Type: *G. striatum* DC.

Mesoperidium very well developed, formed by both abundant hyphae and crystalline matter. Peristome sulcate. Basidiospores 4.0–7.0 µm diam.

Included species. – *Geastrum glaucescens* Speg. agg., *G. parvistriatum* J.C.Zamora & Calonge, *G. pectinatum* Pers., *G. striatum* DC.

Discussion. – This group has mostly been considered a natural group (De Toni, 1887a, as “sect. *Striati*”; Staněk, 1958, as “stirps *Pectinata*”; Ponce de León, 1968, as “group *Pectinata*”; Dörfelt, 1985, as “subg. *Pectinata*”), but often included *G. elegans* and *G. schmidelii* which are placed elsewhere according to our analyses (Fig. 2). Basidium morphology and mesoperidial development justify this separation.

8. *Geastrum* sect. *Hariotia* J.C.Zamora, **sect. nov.** – Type: *G. hariotii* Lloyd (Fig. 3D).

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Rhizomorphs with bipyramidal crystals, isolated or forming rose-like aggregates. Exoperidium saccate to arched and not hygrometric. Mycelial layer encrusting debris; double-layered, with an outer layer formed by skeletal hyphae, and an inner layer formed by generative hyphae. Endoperidial body sessile or shortly stalked. Endoperidial surface rough due to obscure ridges or warts, rarely almost smooth. Mesoperidium almost absent. Peristome sulcate. Basidiospores 3.0–5.0 µm diam., inconspicuously ornamented (ornamentation 0.2–0.4 µm high), and baculate-pilae. Syringaldazine reaction unknown.

Included species. – *Geastrum hariotii* Lloyd agg.

Discussion. – Morphologically homogeneous, species of this group could be confused with those of sect. *Campestris*, but differ in basidiospore features (see above).

9. *Geastrum* sect. *Hieronymia* J.C.Zamora, **sect. nov.** – Type: *G. hieronymi* Henn. (Fig. 3B).

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Rhizomorphs with bipyramidal crystals, often forming rose-like aggregates, sometimes isolated. Exoperidium arched, not hygrometric. Mycelial layer encrusting debris; double-layered, with an outer layer formed by skeletal hyphae, and an inner layer formed by generative hyphae. Endoperidial body stalked. Endoperidial surface spiny, covered with forked processes, formed by bundles of hyphae. Mesoperidium almost absent. Peristome fibrillose, indistinctly delimited. Basidiospores 5.0–7.0 µm diam. Syringaldazine reaction unknown.

Included species. – *Geastrum hieronymi* Henn.

Discussion. – The MP analysis recovered an unsupported clade (data not shown) formed by *G. hieronymi* and *G. entomophilum*. This clade may be a natural group, sharing an unusual endoperidial surface covered by forked processes, formed by bundles of hyphae (Fig. 3B), and may represent another section. However, as noted in the results, contradictory topologies were found in the ML and Bayesian analyses. The ML and Bayesian trees showed an isolated *G. hieronymi*, but group *G. entomophilum* as sister of sect. *Hariotia*, with high statistical support in the Bayesian analysis (MLbs = 67%, pp = 1.0; Fig. 2). Irrespective of the fibrillose peristome, the inclusion of *G. entomophilum* in sect. *Hariotia* and its separation from *G. hieronymi* could be supported by the almost sessile endoperidial body and relatively small basidiospores of *G. entomophilum*, compared with the clearly stalked endoperidial body and bigger basidiospores of *G. hieronymi*. Since *G. hieronymi* does not fit morphologically well in any of the other sections, and its isolation is supported by all phylogenetic analyses, we separate it as its own sect. *Hieronymia*, leaving *G. entomophilum* currently unplaced.

10. *Geastrum* sect. *Myceliostroma* (Henn.) P.Ponce de León in Fieldiana, Bot. 31: 309. 1968. ≡ *Geastrum* subg. *Myceliostroma* Henn. in Hedwigia 43: 185. 1904 – Type: *G. juaruense* Henn.

Rhizomorphs with prism-shaped crystals, covering cystidoid elements, rarely with some horn-like or bipyramidal crystals intermixed. Exoperidium saccate, not hygrometric. Mycelial layer not encrusting debris; double-layered, with an outer layer formed by skeletal hyphae, and an inner layer formed by generative hyphae. Endoperidial body sessile. Endoperidial surface glabrous or with some thick-walled, protruding hyphae. Mesoperidium poorly developed to almost absent. Peristome fibrillose, distinctly delimited and thickened. Basidiospores 3.0–6.0 µm diam., baculate-pilae. Pseudoparenchymatous layer clearly reacting with syringaldazine.

Discussion. – Some species in this section (e.g., those of the *G. velutinum* group) are morphologically similar to some species of sect. *Exareolata* (e.g., *G. argentinum* and *G. albo-*

1 Nomenclature Editor's Note: Under the current wording of Art. 22.2 (McNeill & al., 2012), *Geastrum* subsect. *Sulcostomata* V.J.Staněk cannot be taken as validly published, since the name was published for a subdivision of a genus that included the type (*G. pectinatum*) of that genus. Zamora's (2014) proposal to conserve the type of *Geastrum* using the type of *G. coronatum* has no effect on the invalidly published status of *Geastrum* subsect. *Sulcostomata* because Art. 22.2 does not address conservation. Despite its invalidly published status, *Geastrum* subsect. *Sulcostomata* is used here pending potential amendments to be proposed to Art. 22.2 that will address this situation. — G.M.

nigrum), but rhizomorph crystals and the delimitation of the peristome allow a clear separation of the two groups. It was considered a natural group first by Dissing & Lange (1962), and corresponds exactly to subg. *Myceliostroma* Henn. (Hennings, 1904), incorporated in Ponce de León's (1968) classification, and later accepted by Dörfelt (1985). We consider Ponce de León's sect. *Myceliostroma* as a name at new rank based on Art. 41.5 of the *Code*, since the basionym, *Geastrum* subg. *Myceliostroma* Henn., is indicated and a direct and full reference to its author and place of valid publication, with page reference and date, is given at the end of Ponce de León's paper. In the same work Ponce de León published *Geastrum* sect. *Subepigaea* with a valid Latin diagnosis, selecting *G. javanicum* Lév. as type, another species that could be thought to be part of this group. However, since *G. javanicum* is a rather "enigmatic" species for which we do not know several important morphological traits, such as the delimitation of the peristome, we cannot decide whether sect. *Subepigaea* might correspond to our concept of sect. *Myceliostroma* or to sect. *Exareolata* instead (see above). This is why we decided to recover Ponce's sect. *Myceliostroma* instead of using the doubtful name sect. *Subepigaea*.

10a. *Geastrum* subsect. *Epigaea* Dissing & M.Lange in Bull. Jard. Bot. État Bruxelles 32: 366. 1962 – Type: *G. mirabile* Mont.

= *Coilomyces* Berk. & M.A.Curtis (see *Geastrum* entry, p. 486).

Lignicolous or, less frequently, humicolous species, forming a well-developed subiculum. Basidiomata often caespitose, small to medium-sized. Rhizomorph cystidioid elements narrowly cylindrical, covered with coarse prism-shaped crystals.

Included species. – **Geastrum congolense* Dissing & M.Lange, **G. juruense* Henn., *G. liloi* L.S.Domínguez, *G. pleosporum* Douanla-Meli, *G. schweinitzii* (Berk. & M.A.Curtis) Zeller agg. (including *G. mirabile* Mont., *G. subiculosum* Cooke & Masee, etc.), **G. trichiferum* Rick.

10b. *Geastrum* subsect. *Velutina* J.C.Zamora, **subsect. nov.** – Type: *G. velutinum* Morgan (Fig. 1L).

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Humicolous species with a well-developed rhizomorph system. Basidiomata not caespitose, medium-sized. Rhizomorph cystidioid elements well-developed, more or less vesiculose, and covered with narrow prism-shaped crystals.

Included species. – **Geastrum taylorii* (Lloyd) Jeppson & E.Larss., *G. velutinum* Morgan agg.

11. *Geastrum* sect. *Papillata* De Toni in Rev. Mycol. (Toulouse) 9(34): 65. 1887 ("Papillati") – **Type (designated here):** *G. floriforme* Vittad. (Fig. 1H).

Rhizomorphs with bipyramidal crystals forming rose-like aggregates. Exoperidium saccate to almost planar or somewhat arched, strongly hygrometric. Mycelial layer encrusting debris; double-layered, with an outer layer formed by skeletal hyphae, and an inner layer formed by generative hyphae. Endoperidial body sessile. Endoperidial surface rough due to low ridges or warts. Mesoperidium almost absent. Peristome fibrillose,

indistinctly delimited. Basidiospores 5.0–7.0 µm diam., verrucose to irregularly baculate. Syringaldazine reaction unknown.

Included species. – *Geastrum floriforme* Vittad.

Discussion. – Staněk (1958) considered *G. hungaricum* close to *G. floriforme*, but without molecular data we prefer to be cautious and not to include *G. hungaricum* here, since the endoperidial surface, mesoperidium, and basidiospore features of *G. hungaricum* are more similar to sect. *Geastrum* than to *G. floriforme*. The molecular analyses of Jeppson & al. (2013) also suggest the hypothesis that *G. hungaricum* is probably placed in sect. *Geastrum*.

12. *Geastrum* sect. *Pseudolimbata* J.C.Zamora, **sect. nov.** – Type: *G. pseudolimbatum* Hollós.

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Rhizomorphs with bipyramidal crystals, isolated or forming rose-like aggregates. Exoperidium arched, hygrometric. Mycelial layer encrusting debris; double-layered, with an outer layer formed by skeletal hyphae, and an inner layer formed by generative hyphae. Endoperidial body stalked. Endoperidial surface rough due to low ridges or warts. Mesoperidium almost absent. Peristome fibrillose, distinctly delimited. Basidiospores 5.0–7.0 µm diam., irregularly baculate. Syringaldazine reaction unknown.

Included species. – *Geastrum pseudolimbatum* Hollós.

Discussion. – Although *G. pseudolimbatum* has been considered close to *G. coronatum* (Staněk, 1958), and has even been treated as a mere form of it (Dörfelt & Müller-Uri, 1983), the rough endoperidial surface, the almost complete absence of mesoperidium, and stouter basidia support its separation not only from *G. coronatum*, but also from all other species of sect. *Geastrum*. For these reasons, and without obvious phylogenetic relationships to other morphologically similar species, we separate it as a new section.

13. *Geastrum* sect. *Schmidelia* J.C.Zamora, **sect. nov.** – Type: *G. schmidelii* Vittad.

MycoBank MB 807744

Rhizomorphs with bipyramidal crystals, isolated or forming rose-like aggregates. Exoperidium arched, not hygrometric. Mycelial layer encrusting debris; double-layered, with an outer layer formed by skeletal hyphae, and an inner layer formed by generative hyphae. Endoperidial body more or less stalked. Endoperidial surface glabrous. Mesoperidium almost absent. Peristome sulcate. Basidia rounded, ellipsoid or clavate, stout. Basidiospores 4.0–7.0 µm diam., verrucose to irregularly baculate. Pseudoparenchymatous layer not reacting with syringaldazine.

Included species. – *Geastrum schmidelii* Vittad.

Discussion. – Species in this section are morphologically similar to those of subsect. *Sulcostomata*, but the very poorly developed mesoperidium and less elongated basidia clearly differentiate them (see above).

14. *Geastrum* sect. *Trichaster* (Czern.) P.Ponce de León in Fieldiana, Bot. 31: 310. 1968 ≡ *Trichaster* Czern. in Bull. Soc. Imp. Naturalistes Moscou 18(2): 149. 1845 ≡ *Geastrum*

subg. *Trichaster* (Czern.) P.Ponce de León in Fieldiana, Bot. 31: 310. 1968 ≡ *Geastrum* subsect. *Trichaster* (Czern.) Dissing & M.Lange in Bull. Jard. Bot. État Bruxelles 32: 367. 1962 – Type: *Trichaster melanocephalus* Czern. (≡ *Geastrum melanocephalum* (Czern.) V.J.Staněk, Fig. 11). = *Radiigera* Zeller (see *Geastrum* entry, p. 486).

Rhizomorphs with bipyramidal crystals forming rose-like aggregates, rarely isolated. Exoperidium saccate, arched or pseudoformicate, not hygrometric. Mycelial layer not encrusting debris; single-layered, consisting on generative hyphae. Endoperidial body sessile or stalked. Endoperidial surface glabrous. Mesoperidium almost absent. Peristome fibrillose (sometimes indistinct), mostly delimited. Basidiospores 4.0–6.5 µm diam., baculate-pilate. Pseudoparenchymatous layer reacting with syringaldazine.

Included species. – *Geastrum fuscogleba* (Zeller) Jeppson & E.Larss., *G. melanocephalum* (Czern.) V.J.Staněk, *G. michelianum* Berk. & Broome (North American–European “*G. triplex*”), *G. aff. michelianum* (South America), *G. smardae* V.J.Staněk, *G. triplex* sensu auct. plur. p.p. (Pan-Pacific/Meso-African “*G. triplex*”).

Discussion. – Species of subsect. *Lageniformia* can be confused with those of this section, but both syringaldazine reaction on the pseudoparenchymatous layer and rhizomorph crystals allow their distinction (Zamora & al., 2013). We consider sect. *Trichaster* a natural and rather “intuitive” group, resulting from the combination of stirps “*Smardae*” and “*Triplex*” pro parte of Staněk (1958), and including the type species of *Radiigera* (*R. fuscogleba*). Taking into account that the name *G. triplex* has been applied to several species (Kasuya & al., 2012), and that the next oldest name available for a species in this group is *G. melanocephalum*, the name sect. *Trichaster*, first used by Ponce de León (1968), can be used when expanding its definition to include all species with a mycelial layer of the *Trichaster*-type. We consider sect. *Trichaster* as a name at new rank (Art. 6.10) even if Ponce de León (1968) used the statement “sect. nov.” and provided a Latin diagnosis, since he used the same name and cited the same type that of the genus *Trichaster*. We avoid the use of the names sect. *Cupulati* (De Toni, 1887a) and sect. *Basimyceliata* (Staněk, 1958), which both have *G. triplex* as type, since the “true” *G. triplex* is probably not closely related to any species of sect. *Trichaster* as defined here (Kasuya & al., 2012).

Species incertae sedis (those treated in Sunhede, 1989 as different species, or published after him): *G. entomophilum* Fazolino, Calonge & Baseia, **G. hungaricum* Hollós, *G. reticulatum* Desjardin & Hemmes.

Comments on recent combinations of *Radiigera* in *Geastrum* from Jeppson & al. (2013)

Geastrum flexuosum (L.S.Domínguez & Castellano) Jeppson & E.Larss. in Syst. Biodivers. 11(4): 445. 2013 ≡ *Radiigera flexuosa* L.S.Domínguez & Castellano in Mycologia 88: 877. 1996 – Holotype: SWEDEN. Stockholm province: Uppland, Torsovi parish (Enköping district), Röllingen Island, 16 Sep 1974, *L. Kers* 4548 (UPS!).

Discussion. – Both molecular phylogenetic data of a recently collected specimen from the same place as the holotype, and morphological traits, support the inclusion of this species in the rather complex *G. saccatum* s.l., treated here as the paraphyletic subsect. *Marginata*.

Geastrum fuscogleba (Zeller) Jeppson & E.Larss. in Syst. Biodivers. 11: 445. 2013 ≡ *Radiigera fuscogleba* Zeller in Mycologia 36: 633. 1944 – Holotype: U.S.A. Oregon: Multnomah Co., Mount Scott, Jul 1943, *W. Gruber* 8 (NY!).

Discussion. – Basidiomata from the holotype and other specimens showed a rather thick but single-layered mycelial layer, made up exclusively of generative hyphae, and rose-like aggregates of bipyramidal crystals on the rhizomorphs. These are morphological traits only shared by species of *Geastrum* sect. *Trichaster*. Molecular data strongly support (MPbs = 100%, MLbs = 100%, pp = 1.0) the inclusion of *G. fuscogleba* in that section as well. Jeppson & al. (2013) did not study any specimen of this taxon, but used the GenBank sequences DQ218609 (specimen voucher OSC58979) and DQ219234 (same voucher) analysed in Hosaka & al. (2007). The study of the type specimen, which allowed us to confirm that the diagnostic morphological characters used here support the phylogenetic position of the species, was crucial for our work, since *R. fuscogleba* is the type of the genus *Radiigera*.

Geastrum taylorii (Lloyd) Jeppson & E.Larss. in Syst. Biodivers. 11: 445. 2013 ≡ *Mesophelia taylorii* Lloyd in Mycol. Writings 7(73): 1305. 1924 ≡ *Radiigera taylorii* (Lloyd) Zeller in Mycologia 36: 636. 1944 – Holotype: U.S.A. California: Humboldt County, Eureka, 1924, *W. Taylor* (NY!).

Discussion. – After studying the holotype and some other collections, we found medium-sized basidiomata, with both a double mycelial layer with a felted outer surface, and rhizomorphs with well-developed cystidioid cells covered with thin prism-shaped crystals. These characters support the inclusion of this species in *Geastrum* subsect. *Velutina*. Jeppson & al. (2013), using GenBank sequences of specimens not morphologically studied by them, also showed *R. taylorii* nested in *Geastrum*.

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■ LITERATURE CITED

- Binder, M. & Hibbet, D.S. 2006. Molecular systematics and biological diversification of Boletales. *Mycologia* 98: 971–981. <http://dx.doi.org/10.3852/mycologia.98.6.971>
- Borchsenius F. 2007. FastGap, version 1.0.8. Software distributed by the author. http://192.38.46.42/aubot/fb/FastGap_home.htm (accessed 1 Nov 2012).
- Bronchart, R., Calonge, F.D. & Demoulin, V. 1975. Nouvelle contribution à l'étude de l'ultrastructure de la paroi sporale des Gastéromycètes. *Bull. Soc. Mycol. France* 91: 231–246.
- Brown, J.M., Hedtke, S.M., Lemmon, A.R. & Lemmon, E.M. 2010. When trees grow too long: Investigating the causes of highly inaccurate Bayesian branch-length estimates. *Syst. Biol.* 59: 145–161. <http://dx.doi.org/10.1093/sysbio/syp081>
- Calendini, F. & Martin, J.-F. 2005. PaupUp, version 1.0.3.1: A free graphical frontend for PAUP* DOS software. <http://www.agro-montpellier.fr/sppe/Recherche/JFM/PaupUp> (accessed 1 Jul 2012).
- Calonge, F.D. 1998. Gasteromycetes. I. Lycoperdales, Nidulariales, Phallales, Sclerodermatales, Tulostomales. *Fl. Mycol. Iber.* 3: 1–271.
- Calonge, F.D. & Martín, M.P. 2003. *Zelleromyces hispanicus*, the gasteroid phase of *Lactarius aurantiacus*. *Bol. Soc. Micol. Madrid* 27: 231–236.
- Castresana, J. 2000. Selection of conserved blocks from multiple alignments for their use in phylogenetic analysis. *Molec. Biol. Evol.* 17: 540–552. <http://dx.doi.org/10.1093/oxfordjournals.molbev.a026334>
- Clements, F.E. & Shear, C.L. 1931. *The genera of fungi*. New York: Wilson.
- Coker, W.C. 1924. The geasters of the United States and Canada. *J. Elisha Mitchell Sci. Soc.* 39: 170–224.
- Cunningham, G.H. 1944 [reprint 1979]. The Gasteromycetes of Australia and New Zealand. *Biblioth. Mycol.* 67: 1–239.
- Da Silva, B.D.B., Cabral, T.S., Marihno, P., Ishikawa, N.K. & Baseia, I.G. 2013. Two new species of *Geastrum* (Geastraceae, Basidiomycota) found in Brazil. *Nova Hedwigia* 96: 445–456. <http://dx.doi.org/10.1127/0029-5035/2013/0089>
- Darriba, D., Taboada, G.L., Doallo, R. & Posada, D. 2012. jModelTest 2: More models, new heuristics and parallel computing. *Nature Meth.* 9: 772. <http://dx.doi.org/10.1038/nmeth.2109>
- De Toni, G.B. 1887a. Revisio monographica generis Geasteris Mich. e tribu Gasteromycetum. *Rev. Mycol. (Toulouse)* 9(34): 61–77.
- De Toni, G.B. 1887b. Revisio monographica generis Geasteris Mich. e tribu Gasteromycetum. *Rev. Mycol. (Toulouse)* 9(35): 125–133.
- Demoulin, V. 1984. Typification of *Geastrum* Pers.: Pers. and its orthographic variant *Geaster* (Gasteromycetes). *Taxon* 33: 498–501. <http://dx.doi.org/10.2307/1220990>
- Dissing, H. & Lange, M. 1961. The genus *Geastrum* in Denmark. *Bot. Tidsskr.* 57: 1–27.
- Dissing, H. & Lange, M. 1962. Gasteromycetes of Congo. *Bull. Jard. Bot. État Bruxelles* 32: 325–416.
- Domínguez de Toledo, L.S. & Castellano, M.A. 1998. A revision of the genera *Radiigera* and *Pyrenogaster*. *Mycologia* 88: 863–884. <http://dx.doi.org/10.2307/3760983>
- Dörfelt, H. 1985. *Die Erdsterne: Geastraceae und Astraeaceae*. Wittenberg Lutherstadt: Ziemsen.
- Dörfelt, H. & Müller-Uri, C. 1983. Notizen zur Systematik der Geastrales. *Boletus* 7: 13–20.
- Douanla-Meli, C., Langer, E. & Calonge, F.D. 2005. *Geastrum pleosporus* sp. nov., a new species of Geastraceae identified by morphological and molecular phylogenetic data. *Mycol. Progr.* 4: 239–250. <http://dx.doi.org/10.1007/s11557-006-0127-3>
- Eberhardt, U. & Verbeken, A. 2004. Sequestrate *Lactarius* species from tropical Africa: *L. angiocarpus* sp. nov. and *L. dolichocaulis* comb. nov. *Mycol. Res.* 108: 1042–1052. <http://dx.doi.org/10.1017/S0953756204000784>
- Felsenstein, J. 1985. Confidence limits on phylogenies: An approach using the bootstrap. *Evolution* 39: 783–791. <http://dx.doi.org/10.2307/2408678>
- Fischer, E. 1933. *Unterklasse Eubasidiï, Reihe Gastromycetee*. Vol. 7a of: Engler, A. & Prantl, K (eds.), *Die natürlichen Pflanzenfamilien*, ed. 2. Leipzig: Engelmann.
- Fries, E. 1829. *Systema mycologicum*, vol. 3. Greifswald: sumtibus Ernesti Maurittii.
- Gardes, M. & Bruns, T.D. 1993. ITS primers with enhanced specificity for basidiomycetes: Application to the identification of mycorrhizae and rusts. *Molec. Ecol.* 2: 113–118. <http://dx.doi.org/10.1111/j.1365-294X.1993.tb00005.x>
- Hall, T.A. 1999. BioEdit: A user-friendly biological sequence alignment editor and analysis program for Windows 95/98/NT. *Nucl. Acids Symp. Ser.* 41: 95–98.
- Hemmes, D.E. & Desjardin, D.E. 2011. Earthstars (*Geastrum*, *Myriostoma*) of the Hawaiian Islands including two new species, *Geastrum litchiforme* and *Geastrum reticulatum*. *Pacific Sci.* 65: 477–496. <http://dx.doi.org/10.2984/65.4.477>
- Hennings, P. 1904. Fungi amazonici I. a cl. Ernesto Ule collecti. *Hedwigia* 43: 154–186.
- Hibbet, D.S. 2007. After the gold rush, or before the flood? Evolutionary morphology of mushroom-forming fungi (Agaricomycetes) in the early 21st century. *Mycol. Res.* 111: 1001–1018. <http://dx.doi.org/10.1016/j.mycres.2007.01.012>
- Hillis, D.M. & Bull, J.J. 1993. An empirical test of bootstrapping as a method for assessing confidence in phylogenetic analyses. *Syst. Biol.* 42: 182–192. <http://dx.doi.org/10.1093/sysbio/42.2.182>
- Hollós, L. 1903. *Magyarország Gasteromycetái*. Budapest: Franklin-Társulat.
- Hörandl, E. & Stuessy, T.F. 2010. Paraphyletic groups as natural units of biological classification. *Taxon* 59: 1641–1653.
- Hosaka, K. & Castellano, M.A. 2008. Molecular phylogenetics of Geastrales with special emphasis on the position of *Sclerogaster*. *Bull. Natl. Sci. Mus. Tokyo, B.* 34: 161–173.
- Hosaka, K., Bates, S.T., Beever, R.E., Castellano, M.A., Colgan, W., Domínguez, L.S., Nouhra, E.R., Geml, J., Giachini, A.J., Kenney, S.R., Simpson, N.B., Spatafora, J.W. & Trappe, J.M. 2007 (“2006”). Molecular phylogenetics on the gomphoid-phalloid fungi with an establishment of the new subclass Phallomycetidae and two new orders. *Mycologia* 98: 949–959. <http://dx.doi.org/10.3852/mycologia.98.6.949>
- Jeppson, M. 2013. Jordstjärnor. *Mykol. Publ.* 6: 1–228.
- Jeppson, M., Nilsson, R.H. & Larsson, E. 2013. European earthstars in Geastraceae (Geastrales, Phallomycetidae): A systematic approach using morphology and molecular sequence data. *Syst. Biodivers.* 11: 437–465. <http://dx.doi.org/10.1080/14772000.2013.857367>
- Kasuya, T., Hosaka, K., Uno, K. & Kakishima, M. 2012. Phylogenetic placement of *Geastrum melanocephalum* and polyphyly of *Geastrum triplex*. *Mycoscience* 53: 411–426. <http://dx.doi.org/10.1007/S10267-012-0186-Z>
- Katoh, K., Misawa, K., Kuma, K. & Miyata, T. 2002. MAFFT: A novel method for rapid multiple sequence alignment based on fast Fourier transform. *Nucl. Acids Res.* 30: 3059–3066. <http://dx.doi.org/10.1093/nar/gk1436>

- Kirk, P.M., Cannon, P.F., Minter, D.W. & Stalpers, J.A. 2008. *Ainsworth and Bisby's Dictionary of the Fungi*, 10th ed. Wallingford, U.K.: CABI Europe.
- Kretzer, A. & Bruns, T.D. 1999. Use of *atp6* in fungal phylogenetics: An example from the Boletales. *Molec. Phylogen. Evol.* 13: 483–492. <http://dx.doi.org/10.1006/mpev.1999.0680>
- Kumar, S., Skjæveland, A., Orr, R.J.S., Enger, P., Ruden, T., Mevik, B.-H., Burki, F., Botnen, A. & Shalchian-Tabrizi, K. 2009. AIR: A batch-oriented web program package for construction of supermatrices ready for phylogenomic analyses. *B. M. C. Bioinf.* 10: 357. <http://dx.doi.org/10.1186/1471-2105-10-357>
- Lewis, P.O. 2001. A likelihood approach to estimating phylogeny from discrete morphological character data. *Syst. Biol.* 50: 913–925. <http://dx.doi.org/10.1080/106351501753462876>
- Lloyd, C.G. 1902. The Geastrae. *Bull. Lloyd Libr. Bot.* 2: 1–43.
- Long, W.H. & Stouffer, D.J. 1948. Studies in the Gasteromycetes: XVI. The Geastraceae of the south-western United States. *Mycologia* 40: 547–585. <http://dx.doi.org/10.2307/3755257>
- Lutzoni, F., Kauff, F., Cox, C.J., McLaughlin, D., Celio, G., Dentinger, B., Padamsee, M., Hibbett, D., James, T.Y., Baloch, E., Grube, M., Reeb, V., Hofstetter, V., Schoch, C., Arnold, E., Miadlikowska, J., Spatafora, J., Johnson, D., Hambleton, S., Crockett, M., Shoemaker, R., Sung, G.-H., Lücking, R., Lumbsch, T., O'Donnell, K., Binder, M., Diederich, P., Ertz, D., Gueidan, C., Hansen, K., Harris, R.C., Hosaka, K., Lim, Y.-W., Matheny, B., Nishida, H., Pfister, D., Rogers, J., Rossman, A., Schmitt, I., Sipman, H., Stone, J., Sugiyama, J., Yahr, R. & Vilgalys, R. 2004. Assembling the fungal Tree of Life: Progress, classification, and evolution of subcellular traits. *Amer. J. Bot.* 91: 1446–1480. <http://dx.doi.org/10.3732/ajb.91.10.1446>
- Marshall, D.C. 2010. Cryptic failure of partitioned Bayesian phylogenetic analyses: Lost in the land of long trees. *Syst. Biol.* 59: 108–117. <http://dx.doi.org/10.1093/sysbio/syp080>
- Martín, M.P. & Winka, K. 2000. Alternative methods of extracting and amplifying DNA from lichens. *Lichenologist* 32: 189–196. <http://dx.doi.org/10.1006/lich.1999.0254>
- Matheny, P.B., Liu, Y.J., Ammirati, J.F. & Hall, B.D. 2002. Using RPB sequences to improve phylogenetic inference among mushrooms (*Inocybe*, Agaricales). *Amer. J. Bot.* 89: 688–698. <http://dx.doi.org/10.3732/ajb.89.4.688>
- McNeill, J., Barrie, F.R., Burdet, H.M., Demoulin, V., Hawksworth, D.L., Marhold, K., Nicolson, D.H., Prado, J., Silva, P.C., Skog, J.E., Wiersema, J.H., & Turland, N.J. 2006. *International Code of Botanical Nomenclature (Vienna Code): Adopted by the Seventeenth International Botanical Congress Vienna, Austria, July 2005*. Regnum Vegetabile 146. Ruggell: Gantner. <http://www.iapt-taxon.org/icbn/main.htm>
- McNeill, J., Barrie, F.R., Buck, W.R., Demoulin, V., Greuter, W., Hawksworth, D.L., Herendeen, P.S., Knapp, S., Marhold, K., Prado, J., Prud'homme van Reine, W.F., Smith, G.F., Wiersema, J.H. & Turland, N.J. 2012. *International Code of Nomenclature for algae, fungi, and plants (Melbourne Code): Adopted by the Eighteenth International Botanical Congress Melbourne, Australia, July 2011*. Regnum Vegetabile 154. Königstein: Koeltz Scientific Books. <http://www.iapt-taxon.org/nomen/main.php>
- Meseguer, A.S., Aldasoro, J.S. & Sanmartín, I. 2013. Bayesian inference of phylogeny, morphology and range evolution reveals a complex evolutionary history in St. John's wort (*Hypericum*). *Molec. Phylogen. Evol.* 67: 379–403. <http://dx.doi.org/10.1016/j.ympev.2013.02.007>
- Morgan, A.P. 1884. The North American Geasters. *Amer. Naturalist* 18: 963–970. <http://dx.doi.org/10.1086/273780>
- Morgan, A.P. 1889. North American Fungi. Second paper. The Gasteromycetes. Order II.—Lycoperdaceae. *J. Cincinnati Soc. Nat. Hist.* 12: 8–22.
- Orihara, T., Kasuya, T., Phongpaichit, S. & Dissara, Y. 2008. *Radiigera tropica* (Geastraceae, Geastrales), a new species from a tropical rain forest of Thailand. *Mycotaxon* 105: 111–117.
- Peintner, U., Bougher, N.L., Castellano, M.A., Moncalvo, J.M., Moser, M.M., Trappe, J.M. & Vilgalys, R. 2001. Multiple origins of sequestrate fungi related to *Cortinarius* (Cortinariaceae). *Amer. J. Bot.* 88: 2168–2179. <http://dx.doi.org/10.2307/3558378>
- Pérez-Silva, E., Herrera, T. & Esqueda-Valle, M. 1999. Species of *Geastrum* (Basidiomycotina: Geastraceae) from Mexico. *Revista Mex. Micol.* 15: 89–104.
- Persoon, C.H. 1794. Neuer Versuch einer systematischen Einteilung der Schwämme. *Neues Mag. Bot.* 1: 63–128.
- Persoon, C.H. 1801. *Synopsis methodica fungorum*. Göttingen: apud Henricum Dieterich.
- Ponce de León, P. 1968. A revision of the family Geastraceae. *Fieldiana, Bot.* 31: 301–349.
- Rambaut, A. 2007. FigTree, version 1.3. <http://tree.bio.ed.ac.uk/software/figtree/> (accessed: 1 Oct 2012).
- Rambaut, A. & Drummond, A.J. 2007. Tracer, version 1.4. <http://tree.bio.ed.ac.uk/software/tracer/> (accessed: 1 Nov 2012).
- Robert, V., Stegehuis, G. & Stalpers, J. 2005 [continuously updated]. The Mycobank engine and related databases. <http://www.mycobank.org/> (accessed 17 Feb 2013).
- Ronquist, F., Teslenko, M., Van der Mark, P., Ayres, D.L., Darling, A., Höhna, S., Larget, B., Liu, L., Suchard, M.A. & Huelsenbeck, J.P. 2012. MrBayes 3.2: Efficient Bayesian phylogenetic inference and model choice across a large model space. *Syst. Biol.* 61: 539–542. <http://dx.doi.org/10.1093/sysbio/sys029>
- Shimodaira, H. & Hasegawa, M. 1999. Multiple comparisons of log-likelihoods with applications to phylogenetic inference. *Molec. Biol. Evol.* 16: 1114–1116. <http://dx.doi.org/10.1093/oxfordjournals.molbev.a026201>
- Simmons, M.P. & Ochoterena, H. 2000. Gaps as characters in sequence-based phylogenetic analyses. *Syst. Biol.* 49: 369–381. <http://dx.doi.org/10.1093/sysbio/49.2.369>
- Soltis, D.E., Smith, S.A., Cellinese, N., Wurdack, K.J., Tank, D.C., Brockington, S.F., Refulio-Rodriguez, N.F., Walker, J.B., Moore, M.J., Carlswald, B.S., Bell, C.D., Latvis, M., Crawley, S., Black, C., Diouf, D., Xi, Z., Rushworth, C.A., Gitzendanner, M.A., Sytsma, K.J., Qiu, Y.-L., Hilu, K.W., Davis, C.C., Sanderson, M.J., Beaman, R.S., Olmstead, R.G., Judd, W.S., Donoghue, M.J. & Soltis, P.S. 2011. Angiosperm phylogeny: 17 genes, 640 taxa. *Amer. J. Bot.* 98: 704–730. <http://dx.doi.org/10.3732/ajb.1000404>
- Soto, M.K. & Wright, J.E. 2000. Taxonomía del género *Geastrum* (Basidiomycetes, Lycoperdaceae) en la provincia de Buenos Aires, Argentina. *Bol. Soc. Argent. Bot.* 34: 185–201.
- Spezzadini, C. 1881. Fungi Argentinii additis nonnullis brasiliensibus montevidensibusque. *Anales Soc. Argent. Bot.* 12: 241–258.
- Staněk, V.J. 1958. Geastraceae. Pp. 392–526, 777–795 in: Pilát, A. (ed.), *Flora ČSR BI: Gasteromycetes*. Prague: Československá Akademie Věd.
- Stiller, J.W. & Hall, B.D. 1997. The origin of red algae: Implications for plastid evolution. *Proc. Natl. Acad. Sci. U.S.A.* 94: 4520–4525. <http://dx.doi.org/10.1073/pnas.94.9.4520>
- Sunhede, S. 1989. Geastraceae (Basidiomycotina): Morphology, ecology and systematics with special emphasis on the North European species. *Syn. Fungorum* 1: 1–534.
- Swofford, D.L. 2003. PAUP*: Phylogenetic analysis using parsimony (*and other methods), version 4.0b10. Sunderland, Massachusetts: Sinauer.
- Taylor, J.V., Jacobson, D.J., Kroken, S., Kasuga, T., Geiser, D.M., Hibbet, D.S. & Fisher, M.C. 2000. Phylogenetic species recognition and species concepts in Fungi. *Fungal Genet. Biol.* 31: 21–32. <http://dx.doi.org/10.1006/fgbi.2000.1228>
- Vilgalys, R. & Hester, M. 1990. Rapid genetic identification and mapping of enzymatically amplified DNA from several *Cryptococcus* species. *J. Bacteriol.* 172: 4238–4246.
- Vittadini, C. 1842. *Monographia lycoperdineorum*. Torino: ex Officina regia.

- Voss, E.G., Burdet, H.M., Chaloner, W.G., Demoulin, V., Hiepko, P., McNeill, J., Meikle, R.D., Nicolson, D.H., Rollins, R.C., Silva, P.C. & Greuter, W. 1983. *International Code of Botanical Nomenclature: Adopted by the Thirteenth International Botanical Congress, Sydney, August 1981*. Regnum Vegetabile 111. Utrecht: Bohn, Scheltema & Holkema; Boston: W. Junk.
- White, T.J., Bruns, T., Lee, S. & Taylor, J. 1990. Amplification and direct sequencing of fungal ribosomal RNA genes for phylogenetics. Pp. 315–322 in: Innis, M., Gelfand, J., Sninsky, J. & White, T. (eds.), *PCR protocols: A guide to methods and applications*. Orlando: Academic Press.
- Zamora, J.C. 2014. Proposal to conserve the name *Geastrum* (*Geastrales*, *Basidiomycota*) with a conserved type. *Taxon* 63: 679–680. <http://dx.doi.org/10.12705/633.22> [this issue]
- Zamora, J.C., Calonge, F.D. & Martín, M.P. 2013. New sources of taxonomic information for earthstars (*Geastrum*, *Geastraceae*, *Basidiomycota*): Phenoloxidas and rhizomorph crystals. *Phytotaxa* 132: 1–20. <http://dx.doi.org/10.11646/phytotaxa.132.1.1>
- Zhou, T., Chen, Y., Zhao, L., Fu, H. & Yang, B. 2006. *Flora Fungorum Sinicorum*, vol. 36, *Geastraceae—Nidulariaceae*. Beijing: Science Press.
- Zwickl, D.J. 2006. *Genetic algorithm approaches for the phylogenetic analysis of large biological sequence datasets under the maximum likelihood criterion*. Ph.D. dissertation, The University of Texas at Austin.
- Zwickl, D.J. & Hillis, D.M. 2002. Increased taxon sampling greatly reduces phylogenetic error. *Syst. Biol.* 51: 588–598. <http://dx.doi.org/10.1080/10635150290102339>

Appendix 1. Specimens used in this study including taxon name, collection locality, collector, collection no. (only when available; all others are *s.n.*), herbarium number, and GenBank accession numbers for 5.8S, nrLSU, *rpb1*, and *atp6* sequences respectively. A dash (–) indicates that the DNA sequence was not obtained, and an asterisk (*) denotes a single sequence less than 200 bp that is provided below Appendix 1.

Schenella pityophila (Malençon & Rioussel) Estrada & Lado 1, Spain: Albacete, A. Rodríguez (Herb. Zamora 530), KF988346, KF988464, KF988599, KF988734; *Sch. pityophila* 2, Spain: Albacete, A. Rodríguez (Herb. Zamora 531), KF988347, KF988465, KF988600, KF988735; *Myriostoma coliforme* (With.:Pers.) Corda 1, Spain: Madrid, J.C. Zamora, B. Zamora & J.C. Campos (Herb. Zamora 496), KF988337, KF988466, KF988601, KF988736; *M. coliforme* 2, Argentina: Entre Ríos, S. Suaza, J. Maller & J.C. Zamora (MA-Fungi 83759), KF988348, KF988467, KF988602, KF988737; *Geastrum albonigrum* Calonge & Mata, Panama: Isla de Coibas, F. Pando & M.P. Núñez (MA-Fungi 36140-2), KF988349, KF988468, KF988603, KF988738; *G. aff. arenarium* Lloyd 1, Spain: Madrid, F. Prieto (MA-Fungi 68191), KF988350, KF988469, KF988604, KF988739; *G. aff. arenarium* 2, Spain: Madrid, J.C. Zamora (Herb. Zamora 76), KF988338, KF988470, KF988605, KF988740; *G. cf. arenarium*, Argentina: La Rioja, L. Papinutti & G. Rolón (MA-Fungi 83760), KF988351, KF988471, KF988606, KF988741; *G. argentinum* Speg. 1, Argentina: Buenos Aires, F. Kuhar & J.C. Zamora (LPS 48446), KF988352, KF988472, KF988607, KF988742; *G. argentinum* 2, Argentina: Salta, L. Papinutti, G. Rolón & J.C. Zamora (MA-Fungi 82605), KF988353, KF988473, KF988608, KF988743; *G. berkeleyi* Massee 1, Spain: Cuenca, F. Prieto & A. González (MA-Fungi 74668), KF988354, KF988474, KF988609, KF988744; *G. berkeleyi* 2, Sweden: Gotland, S. Sunhede (Herb. Sunhede 7724), KF988355, KF988475, KF988610, KF988745; *G. berkeleyi* 3, Sweden: Gotland, E. Bohus Jensen (Herb. Zamora 504), KF988356, KF988476, KF988611, KF988746; *G. cf. calceum* Lloyd 1, Brazil: Rio Grande do Norte, M.M.B. Barbosa, B. Tomio, T. Lockwood & I.G. Baseia (UFRN-Fungi 723), KF988340, KF988477, KF988612, KF988747; *G. cf. calceum* 2, Argentina: Tucumán, E. Grassi (MA-Fungi 83761), KF988341, KF988478, KF988613, –; *G. campestre* Morgan 1, Spain: Madrid, L. Penelas (Herb. Zamora 283), JN943167, JN939575, JN91286, KF988748; *G. campestre* 2, Sweden: Skåne, S. Sunhede (Herb. Sunhede 7575), KF988357, KF988479, KF988614, KF988749; *G. campestre* 3, U.S.A.: Arizona, J. States (MICH 28566), KF988358, KF988480, KF988615, KF988750; *G. corollinum* (Batsch) Hollós 1, Spain: Madrid, F.D. Calonge (MA-Fungi 5746), KF988359, KF988481, KF988616, KF988751; *G. corollinum* 2, Sweden: Öland, J.C. Zamora & S. Sunhede (Herb. Sunhede 7744), KF988360, KF988482, KF988617, KF988752; *G. coronatum* Pers. 1, Spain: Madrid, B. Zamora & J.C. Zamora (Herb. Zamora 266), KF988361, KF988483, KF988618, KF988753; *G. coronatum* 2, Sweden: Gotland, J.C. Zamora (Herb. Zamora 522), KF988362, KF988484, KF988619, KF988754; *G. coronatum* 3, U.S.A.: Arizona, J. States (MICH 28567), KF988363, KF988485, KF988620, KF988755; *G. aff. coronatum* Pers. 1, U.S.A.: Wisconsin, J.J. Steinke & P. Watson (MICH 72012), KF988364, KF988486, KF988621, KF988756; *G. aff. coronatum* 2, U.S.A.: Wisconsin, J.J. Steinke (MICH 72014), KF988365, KF988487, KF988622, KF988757; *G. elegans* Vittad. 1, Spain: Ávila, F. Colino (Herb. Zamora 189), KF988366, KF988488, KF988623, KF988758; *G. elegans* 2, Sweden: Gotland, E. Bohus Jensen & J.C. Zamora (UPS F-560810), KF988367, KF988489, KF988624, KF988759; *G. entomophilum* Fazolino, Calonge & Baseia, Brazil: Rio Grande do Norte, I. Baseia (MA-Fungi 70785), KF988368, KF988490, KF988625, KF988760; *G. fimbriatum* Fr. 1, Spain: Huesca, F. Pancorbo, F. Palazón & J. Cuesta (Herb. Zamora 234), KF988369, KF988491, KF988626, KF988761; *G. fimbriatum* 2, Sweden: Västergötland, S. Sunhede & J.C. Zamora (Herb. Sunhede 7739), KF988370, KF988492, KF988627, KF988762; *G. flexuosum* (L.S.Domínguez & Castellano) Jeppson & E.Larss., Sweden: Uppland, Lars. E. Kers 4548 (UPS F-119844), KF988371, KF988493, KF988628; *G. floriforme* (Fl.) Vittad. 1, Spain: Cáceres, F.D. Calonge (MA-Fungi 69173), KF988372, KF988494, KF988629, KF988764; *G. floriforme* 2, Spain: Jaén, J.C. Zamora, J. Señoret & B. Zamora (Herb. Zamora 453), KF988373, KF988495, KF988630, KF988765; *G. fornicatum* (Huds.) Hook. 1, Spain: Valladolid, G. Martínez & A. García (Herb. Zamora 255), KF988374, KF988496, KF988631, KF988766; *G. fornicatum* 2, Spain: Lérida, A. González Cuenca (MA-Fungi 30749), KF988375, KF988497, KF988632, KF988767; *G. fuscogleba* (Zeller) Jeppson & E.Larss. 1, U.S.A.: Oregon, Trappe 1071 (NY), KF988376, KF988498, KF988633, KF988768; *G. fuscogleba* 2, U.S.A.: California, Trappe 9500 (NY), KF988377, KF988499, KF988634, KF988769; *G. glaucescens* Speg. 1, Argentina: La Rioja, L. Papinutti & J.C. Zamora (MA-Fungi 83762), KF988378, KF988500, KF988635, KF988770; *G. glaucescens* 2, Argentina: Catamarca, L. Papinutti & J.C. Zamora (MA-Fungi 83763), KF988379, KF988501, KF988636, KF988771; *G. aff. glaucescens* Speg., Argentina: Santiago del Estero, J.C. Zamora, L. Papinutti & G. Rolón (MA-Fungi 83764), KF988380, KF988502, KF988637, KF988772; *G. hariatii* Lloyd 1, Dominican Republic: Puerto Plata, C. Angelini (MA-Fungi 80070), –, KF988503, KF988638, KF988773; *G. hariatii* 2, Argentina: Tucumán, J. Maller (MA-Fungi 83765), KF988381, KF988504, KF988639, KF988774; *G. aff. hariatii* (1)1, Brazil: Rio Grande do Norte (MA-Fungi 78296), KF988382, KF988505, KF988640, KF988775; *G. aff. hariatii* Lloyd (1)2, Brazil: Rio Grande do Norte, I. Ubaia Doce (MA-Fungi 78289), KF988383, KF988506, KF988641, –, *G. aff. hariatii* Lloyd (2), Mozambique: Niassa, Borge Pettersson 2070 (UPS), KF988342, KF988507, KF988642, –, *G. hieronymi* Henn. 1, Argentina: Córdoba, J.C. Zamora, E. García & A. Aramburri (MA-Fungi 83766), KF988384, KF988508, KF988643, KF988776; *G. hieronymi* 2, Argentina: Entre Ríos, L. Papinutti & G. Rolón (MA-Fungi 83767), KF988344, KF988509, KF988644, KF988777; *G. kotlabae* V.J.Staněk 1, Spain: Madrid, P.P. Daniëls (MA-Fungi 39563), KF988385, KF988510, KF988645, KF988778; *G. kotlabae* 2, Spain: Valladolid, S. Pérez Ortega & J.C. Zamora (Herb. Zamora 440), KF988386, KF988511, KF988646, KF988779; *G. aff. kotlabae* V.J.Staněk, Tanzania: T4, Western Province, T. Saarimäki (MA-Fungi 33300), KF988387, KF988512, KF988647, –, *G. lageniforme* Vittad. 1, Spain: Madrid, B. Zamora (MA-Fungi 83770), KF988388, KF988513, KF988648, KF988780; *G. lageniforme* (1), Brazil, E. Fazolino (MA-Fungi 79056), –, KF988515, KF988650, KF988782; *G. aff. lageniforme* (2), Argentina: Entre Ríos, J. Maller & J.C. Zamora (MA-Fungi 83768), KF988389, KF988516, KF988651, KF988783; *G. aff. lageniforme* (3), Niger: Tamou, P.P. Daniëls (COFC Hama 327), KF988390, KF988517, KF988652, KF988784; *G. aff. lageniforme* (4)1, Argentina: Misiones, E. Grassi & J.C. Zamora (MA-Fungi 83770), –, KF988391, KF988518, KF988653, KF988785; *G. aff. lageniforme* (4)2, Argentina: Entre Ríos, S. Suaza & J.C. Zamora (MA-Fungi 83769), KF988392, KF988519, KF988654, KF988786; *G. aff. lageniforme* (5)1, Portugal: Madeira, F.D. Calonge (MA-Fungi 78398), KF988393, KF988520, KF988655, KF988787; *G. aff. lageniforme* (5)2, Spain: Canary Islands, Tenerife, M.A. Ribes (Herb. Ribes 221210-01), KF988394, KF988521, KF988656, KF988788; *G. melanocephalum* (Czern.) V.J.Staněk 1, Spain: Palencia, L. Barrio & A. Paz (Herb. Zamora 34), KF988395, KF988522, KF988657, KF988789; *G. melanocephalum* 2, Sweden: Västergötland, S. Sunhede & J.C. Zamora (Herb. Sunhede 7737), KF988396, KF988523, KF988658, KF988790; *G. michelianum* Berk. & Broome 1, Sweden: Västergötland, S. Sunhede & J.C. Zamora (Herb. Sunhede 7738), KF988397, KF988524, KF988659,

Appendix 1. Continued.

KF988791; *G. michelianum* 2, Spain: Toledo, A. Moreno, J.C. Campos, F. Álvarez & J.C. Zamora (Herb. Zamora 227), KF988398, KF988525, KF988660, KF988792; *G. michelianum* 3, Spain: Canary Islands, Tenerife, M.Á. Ribes (Herb. Ribes 231208-31), –, KF988526, KF988661, KF988793; *G. aff. michelianum*, Argentina: Misiones, E. Grassi & J.C. Zamora (MA-Fungi 83771), KF988399, KF988527, KF988662, KF988794; *G. minimum* Schwein. agg. (1)1, Spain: Madrid, L. Penelas (Herb. Zamora 191), KF988400, KF988528, KF988663, KF988795; *G. minimum* agg. (1)2, Sweden: Öland, S. Sunhede & J.C. Zamora (Herb. Sunhede 7746), KF988401, KF988529, KF988664, KF988796; *G. minimum* agg. (1)3, U.S.A.: Wisconsin, J.J. Steinke (MICH 72010), KF988402, KF988530, KF988665, KF988797; *G. minimum* agg. (2)1, U.S.A.: Arizona, J. States (MICH 28119), KF988403, KF988531, KF988666, KF988798; *G. minimum* agg. (2)2, Spain: Madrid, F.D. Calonge (MA-Fungi 31530), KF988404, KF988532, KF988667, KF988799; *G. minimum* agg. (2)3, Sweden: Gotland, J.C. Zamora, M. Jeppson & M. Lathi (MA-Fungi 86669), KF988405, KF988533, KF988668, KF988800; *G. morganii* Lloyd, Canada: Quebec, R. Lebeuf (Herb. Lebeuf HRL0177), KF988406, KF988534, KF988669, –, *G. aff. morganii* (1)1, Spain: Oviedo, A. Román & E. Rubio (Herb. Zamora 367), KF988407, KF988535, KF988670, KF988801; *G. aff. morganii* (1)2, Spain: Oviedo, A. Román (Herb. Zamora 525), KF988408, KF988536, KF988671, KF988802; *G. aff. morganii* (2), Argentina: Entre Ríos, S. Suaza & J.C. Zamora (MA-Fungi 83772), KF988409, KF988537, KF988672, KF988803; *G. aff. morganii* (3), Argentina: Salta, L. Papinutti, J.C. Zamora & G. Rolón (MA-Fungi 83773), KF988410, KF988538, KF988673, KF988804; *G. ovalisporum* Calonge & Mor.-Arr., Bolivia: Concepción, B. Moreno-Arroyo & J. Gómez (MA-Fungi 47184), KF988411, KF988539, KF988674, KF988805; *G. parvistriatum* J.C. Zamora & Calonge 1, Spain: Madrid, J.C. Zamora (MA-Fungi 69583), JN943160, JN939560, JN991291, KF988806; *G. parvistriatum* 2, Spain: Madrid, B. Zamora (Herb. Zamora 272), JN943162, JN939572, JN991283, KF988807; *G. pectinatum* Pers. 1, Spain: Madrid, J.C. Zamora (Herb. Zamora 252), KF988412, KF988540, KF988675, KF988808; *G. pectinatum* 2, Sweden: Gotland, T.N. Kristiansen & P. Marstad (UPS F-560803), KF988413, KF988541, KF988676, –, *G. pectinatum* 3, Tanzania: Iringa, B. Pettersson, M. Hedrén & S. P. Kibuwa 791 (UPS F-09935 (161483)), KF988414, KF988542, KF988677, KF988809; *G. pectinatum* 4, Argentina: Buenos Aires, F. Kuhar, V. Castiglia, E. Grassi & J.C. Zamora (MA-Fungi 83774), KF988415, KF988543, KF988678, KF988810; *G. pleosporum* Douanla-Meli, Cameroon: Mbalmayo Forest Reserve, C. Douanla-Meli (MA-Fungi 56971), KF988416, KF988544, KF988679, KF988811; *G. pouzarii* V.J. Staněk 1, Czechoslovakia: Bohemia, F. Kotlaba (MA-Fungi 2944), KF988417, KF988545, KF988680, KF988812; *G. pouzarii* 2, Czechoslovakia: Bohemia (Herb. Sunhede 7494), KF988418, KF988546, KF988681, KF988813; *G. pseudolimbium* Hollós 1, Spain: Badajoz, M. Romero (Herb. Zamora 231), KF988419, KF988547, KF988682, KF988814; *G. pseudolimbium* 2, Sweden: Öland, S. Sunhede & J.C. Zamora (UPS F-560804), KF988420, KF988548, KF988683, KF988815; *G. quadrifidum* Pers.: Pers. 1, Spain: Huesca, M. Chiaffi (Herb. Zamora 170), KF988421, KF988549, KF988684, KF988816; *G. quadrifidum* 2, Sweden: Uppland, I. Olariaga & J.C. Zamora (MA-Fungi 86671), KF988422, KF988550, KF988685, KF988817; *G. quadrifidum* 3, U.S.A.: Colorado, R. Fogel F2078 (MICH 72512), KF988423, KF988551, KF988686, KF988818; *G. rufescens* Pers.: Pers. 1, Spain: Huesca, A. Palazón (Herb. Zamora 253), KF988424, KF988552, KF988687, KF988819; *G. rufescens* 2, Spain: Segovia, J.M. Ruiz (Herb. Zamora 274), KF988425, KF988553, KF988688, KF988820; *G. cf. saccatum* Fr. (1)1, Bolivia: Concepción, B. Moreno-Arroyo & J. Gómez (MA-Fungi 47185-2), KF988426, KF988554, KF988689, KF988821; *G. cf. saccatum* (1)2, Argentina: Buenos Aires, G. Rolón, L. Papinutti & J.C. Zamora (MA-Fungi 83775), KF988427, KF988555, KF988690, KF988822; *G. cf. saccatum* (2), Australia: New South Wales-Queensland (Herb. Sunhede 7749), KF988428, KF988556, KF988691, KF988823; *G. cf. saccatum* (3), Argentina: Entre Ríos, S. Suaza & J.C. Zamora (MA-Fungi 83776), –, KF988557, KF988692, KF988824; *G. cf. saccatum* (4), Japan: Honshu, Saitama, G. Thor 21566 (UPS F-530056), KF988428, KF988558, KF988693, KF988825; *G. cf. saccatum* (5), Argentina: Entre Ríos, J. Maller & J.C. Zamora (MA-Fungi 83777), KF988429, KF988559, KF988694, KF988826; *G. cf. saccatum* (6)1, Spain: Valladolid, D. Herrero (Herb. Zamora 260), KF988430, KF988560, KF988695, KF988827; *G. cf. saccatum* (6)2, Spain: Toledo, J. De Esteban, J.C. Zamora & B. Zamora (Herb. Zamora 461), KF988431, KF988561, KF988696, KF988828; *G. cf. saccatum* (7)1, Niger: Tamou, P.P. Daniëls (COFC Hama 343), KF988432, KF988562, KF988697, KF988829; *G. cf. saccatum* (7)2, Argentina: Salta, L. Papinutti, J.C. Zamora & G. Rolón (MA-Fungi 83778), KF988433, KF988563, KF988698, KF988830; *G. schmidelii* Vittad. 1, Spain: Madrid, J.C. Zamora (Herb. Zamora 279), KF988434, KF988564, KF988699, KF988831; *G. schmidelii* 2, Sweden: Öland, S. Sunhede & J.C. Zamora (UPS F-560805), KF988435, KF988565, KF988700, KF988832; *G. cf. schweinitzii* (Berk. & M.A. Curtis) Zeller (1), Papua New Guinea: Mt. Suru, S. Henrik Kylin 1983 30.X (UPS), KF988436, KF988566, KF988701, KF988833; *G. cf. schweinitzii* (2), Argentina: Misiones, E. Grassi (MA-Fungi 83779), KF988437, KF988567, KF988702, KF988834; *G. cf. schweinitzii* (3), Panama: Coibas Island, F. Pando & M.P. Núñez (MA-Fungi 36141), KF988438, KF988568, KF988703, KF988835; *G. cf. schweinitzii* (4), Argentina: Jujuy, G. Rolón, L. Papinutti & J.C. Zamora (MA-Fungi 83780), KF988439, KF988569, KF988704, KF988836; *G. cf. schweinitzii* (5), Fiji Islands: Suva, S. Henrik Kylin 842 (UPS), –, KF988570, KF988705, –, *G. setiferum* Baseia 1, Argentina: Entre Ríos, L. Papinutti & G. Rolón (MA-Fungi 83781), –, KF988571, KF988706, KF988837; *G. setiferum* 2, Argentina: Entre Ríos, S. Suaza & J.C. Zamora (MA-Fungi 83782), –, KF988572, KF988707, KF988838; *G. smardae* V.J. Staněk 1, Canada: Québec, R. Lebeuf (Herb. Lebeuf HRL 0160), KF988440, KF988573, KF988708, KF988839; *G. smardae* 2, Spain: Valencia, A. Puras (Herb. Zamora 527), KF988441, KF988574, KF988709, KF988840; *G. smithii* Lloyd, Argentina: Córdoba, L. Papinutti, J.C. Zamora & G. Rolón (MA-Fungi 83783), KF988442, KF988575, KF988710, KF988841; *G. cf. stipitatum* Solms ex E. Fisch., Brazil: Rio Negro, M.Á. Perez-De-Gregorio (Herb. Zamora 528), KF988443, KF988576, KF988711, *, *G. striatum* DC. 1, Spain: Valladolid, G. Martínez & A. García (Herb. Zamora 257), JN943164, JN939557, JN991288, KF988842; *G. striatum* 2, Sweden: Uppland, J.C. Zamora (MA-Fungi 86672), KF988443, KF988577, KF988712, KF988843; *G. “triplex”* Jungh. agg. 1, Madagascar: Mahajanga, B. Bremer & P. Erixon 4069b-B69b (UPS F-014630 (213863)), KF988444, KF988578, KF988713, KF988844; *G. “triplex”* agg. 2, Argentina: Catamarca, L. Papinutti, J.C. Zamora & G. Rolón (MA-Fungi 83784), KF988445, KF988579, KF988714, KF988845; *G. cf. velutinum* Morgan (1), India: Uttaranchal, D. Bischt (MA-Fungi 73247), –, KF988580, KF988715, KF988846; *G. cf. velutinum* (2)1, Argentina: Entre Ríos, S. Suaza & J.C. Zamora (MA-Fungi 83785), KF988446, KF988581, KF988716, KF988847; *G. cf. velutinum* (2)2, Argentina: Misiones, E. Grassi & J.C. Zamora (MA-Fungi 83786), KF988447, KF988582, KF988717, KF988848; *G. cf. velutinum* (3), Spain: Canary Islands, Tenerife, M.Á. Ribes (Herb. Ribes 311207-62), KF988448, KF988583, KF988718, KF988849; *G. cf. velutinum* (4), Peru: Cuzco, L. Papinutti, G. Rolón & J.C. Zamora (MA-Fungi 83787), KF988449, KF988584, KF988719, KF988850; *G. violaceum* Rick 1, Argentina: Corrientes, L. Papinutti (BAFC 51671), KF988450, KF988585, KF988720, KF988851; *G. violaceum* 2, Argentina: Misiones, E. Grassi & J.C. Zamora (MA-Fungi 82487), KF988451, KF988586, KF988721, KF988852; *Geastrum* sp. (1)1, Argentina: La Rioja, L. Papinutti & J.C. Zamora (MA-Fungi 83788), KF988452, KF988587, KF988722, KF988853; *Geastrum* sp. (1)2, Argentina: La Rioja, L. Papinutti & J.C. Zamora (MA-Fungi 83789), KF988453, KF988588, KF988723, KF988854; *Geastrum* sp. (2)1, Spain: Toledo, J.M. Santos (MA-Fungi 31143), KF988454, KF988589, KF988724, KF988855; *Geastrum* sp. (2)2, Spain: Teruel, E. Suárez (MA-Fungi 37546), KF988455, KF988590, KF988725, KF988856; *Geastrum* sp. (3), Argentina: Misiones, E. Grassi & J.C. Zamora (MA-Fungi 83790), KF988456, KF988591, KF988726, KF988857; *Geastrum* sp. (4), Peru: Cuzco, L. Papinutti, G. Rolón & J.C. Zamora (MA-Fungi 83791), KF988457, KF988592, KF988727, KF988858; *Geastrum* sp. (5)1, Spain: Cáceres, C. Tovar (Herb. Zamora 145), KF988458, KF988593, KF988728, KF988859; *Geastrum* sp. (5)2, Spain: Jaén, J.C. Zamora, J. Señoret & B. Zamora (Herb. Zamora 450), KF988459, KF988594, KF988729, KF988860; *Geastrum* sp. (6), Argentina: Entre Ríos, S. Suaza, J. Maller & J.C. Zamora (MA-Fungi 83792), KF988460, KF988595, KF988730, KF988861; *Geastrum* sp. (7)1, Argentina: La Rioja, L. Papinutti & J.C. Zamora (MA-Fungi 83793), KF988461, KF988596, KF988731, KF988862; *Geastrum* sp. (7)2, Argentina: La Rioja, L. Papinutti, J.C. Zamora & G. Rolón (MA-Fungi 83794), KF988462, KF988597, KF988732, KF988863; *Geastrum* sp. (8), Argentina: Buenos Aires, F. Kuhar, V. Castiglia & J.C. Zamora (MA-Fungi 83795), KF988463, KF988598, KF988733, KF988864.

* Sequence shorter than 200 bp:

Geastrum cf. *stipitatum* Solms ex E. Fisch., Brazil: Rio Negro, M.Á. Perez-De-Gregorio (Herb. Zamora 528), ATP synthase subunit 6 (*atp6*), partial sequence RCTCCTTTATTAGGTTATTTAAATTTAAAGTTAACTAATTTTCTTTATATTTATATAATAGTTTATTTATTTAGTAATCACAATTCATCTCTTATT TAGAGGAGCCTCTCTATAGCTAATAAACTTCATACTAAATTAGTACCTTCT